

SPIRAL TILINGS

MATHEMATICS

A Coloring Book of Radial Geometry

50 Parametric Designs

MathRelaxing

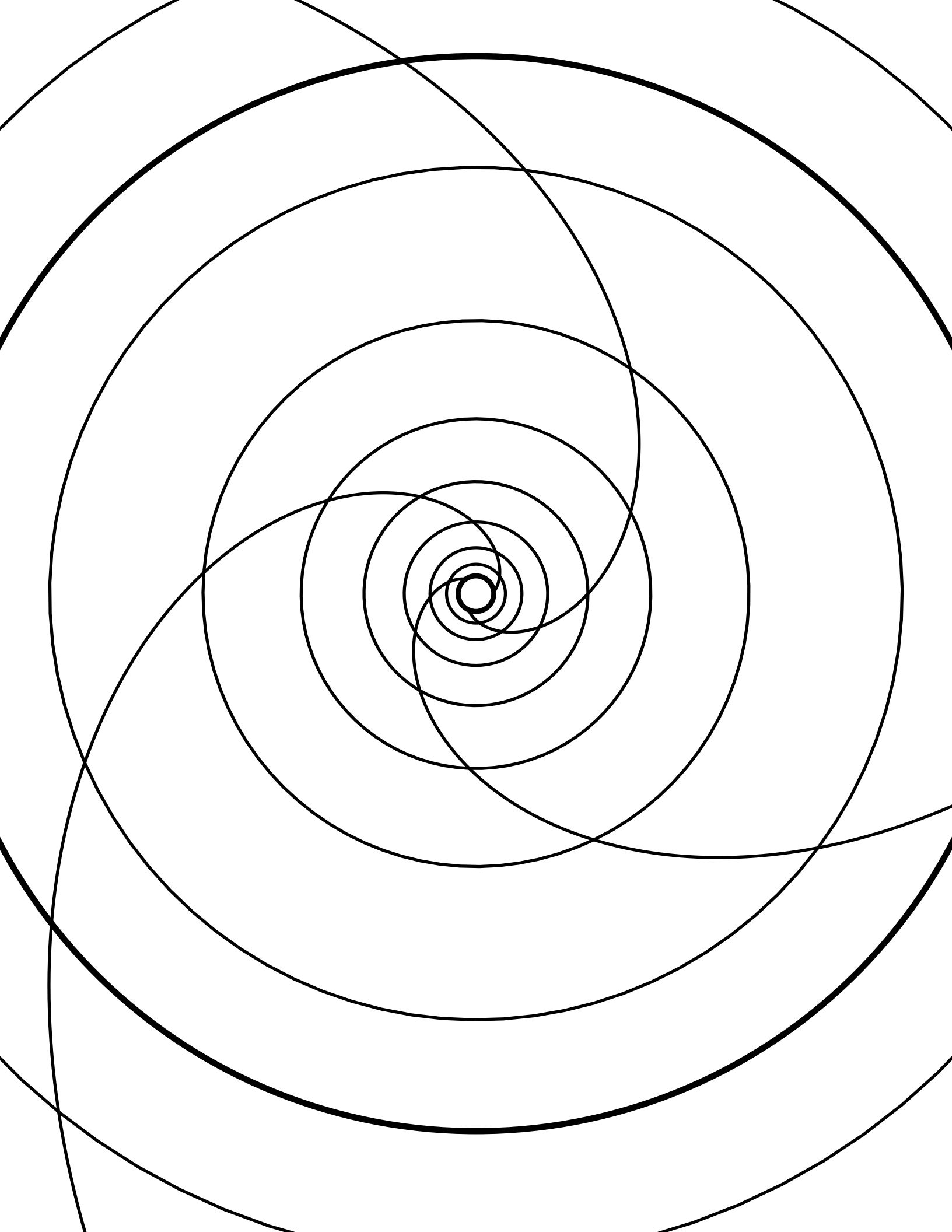
THE MATHEMATICS

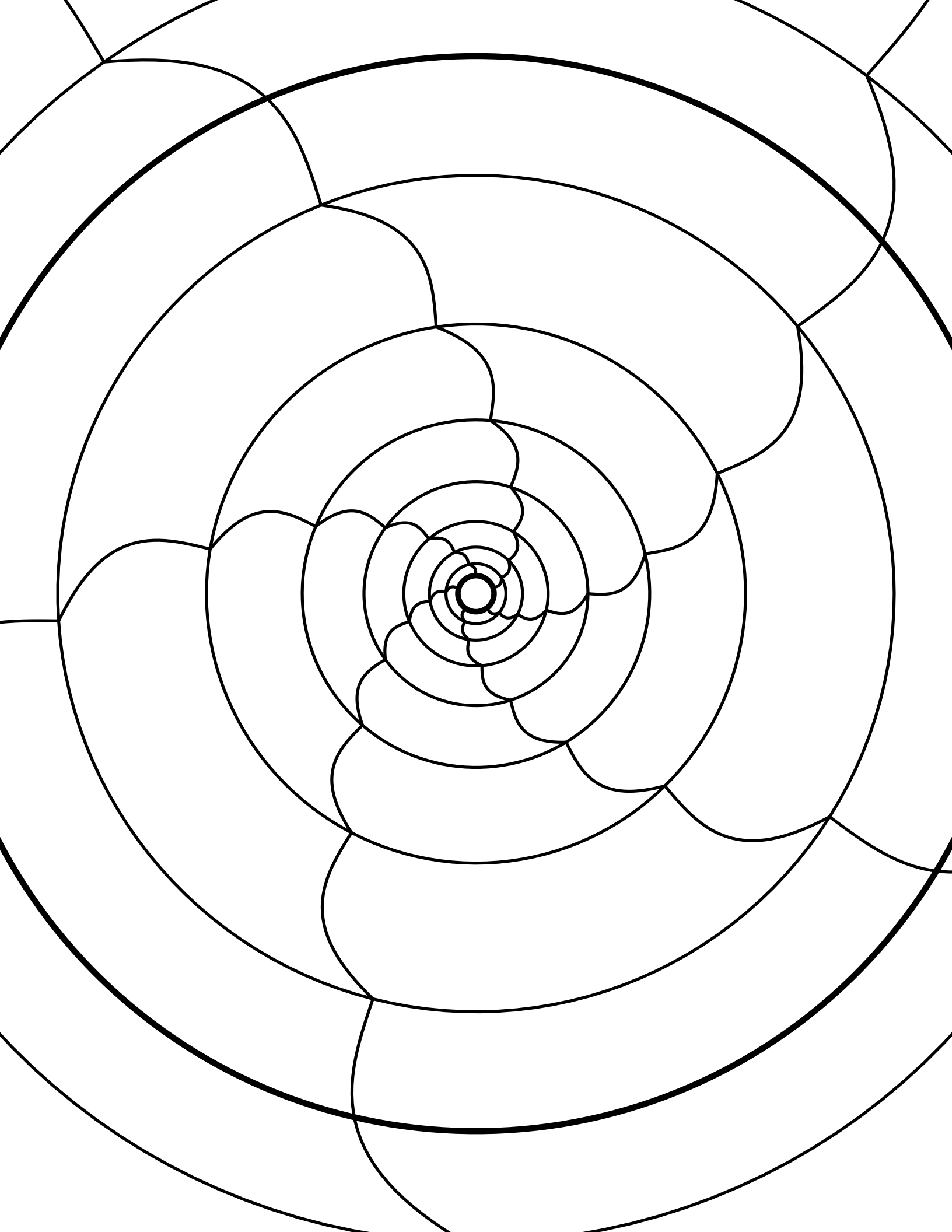
Spiral tilings are a mesmerizing intersection of complex analysis and art. By taking a regular Euclidean grid of squares in the complex plane $\mathbb{C}$ and mapping it through the complex exponential function $f(z) = e^z$, straight lines burst outward into logarithmic spirals of the form $r = ae^{b\theta}$ that endlessly wrap around a central origin.

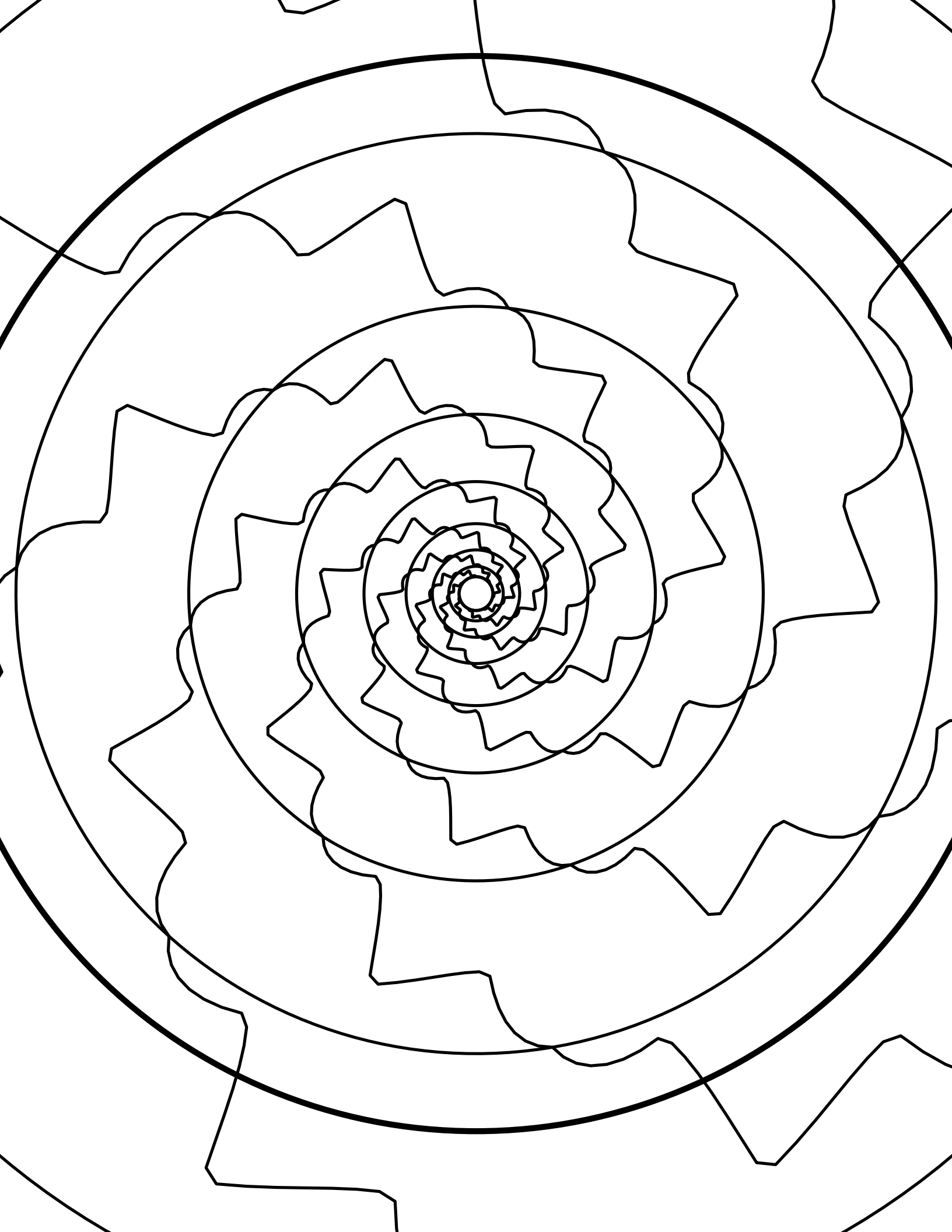
Every design in this book is generated mathematically. By tweaking just three parameters—the number of radial arms N, the radial expansion rate H, and the twist shift Δy—an infinite variety of radial geometries emerge. We then apply periodic deformations (such as sine waves or zig-zags) to the edges in the pre-image grid. When mapped, these become interlocking jigsaw pieces, scales, and floral petals that fit together perfectly without a single gap or overlap.

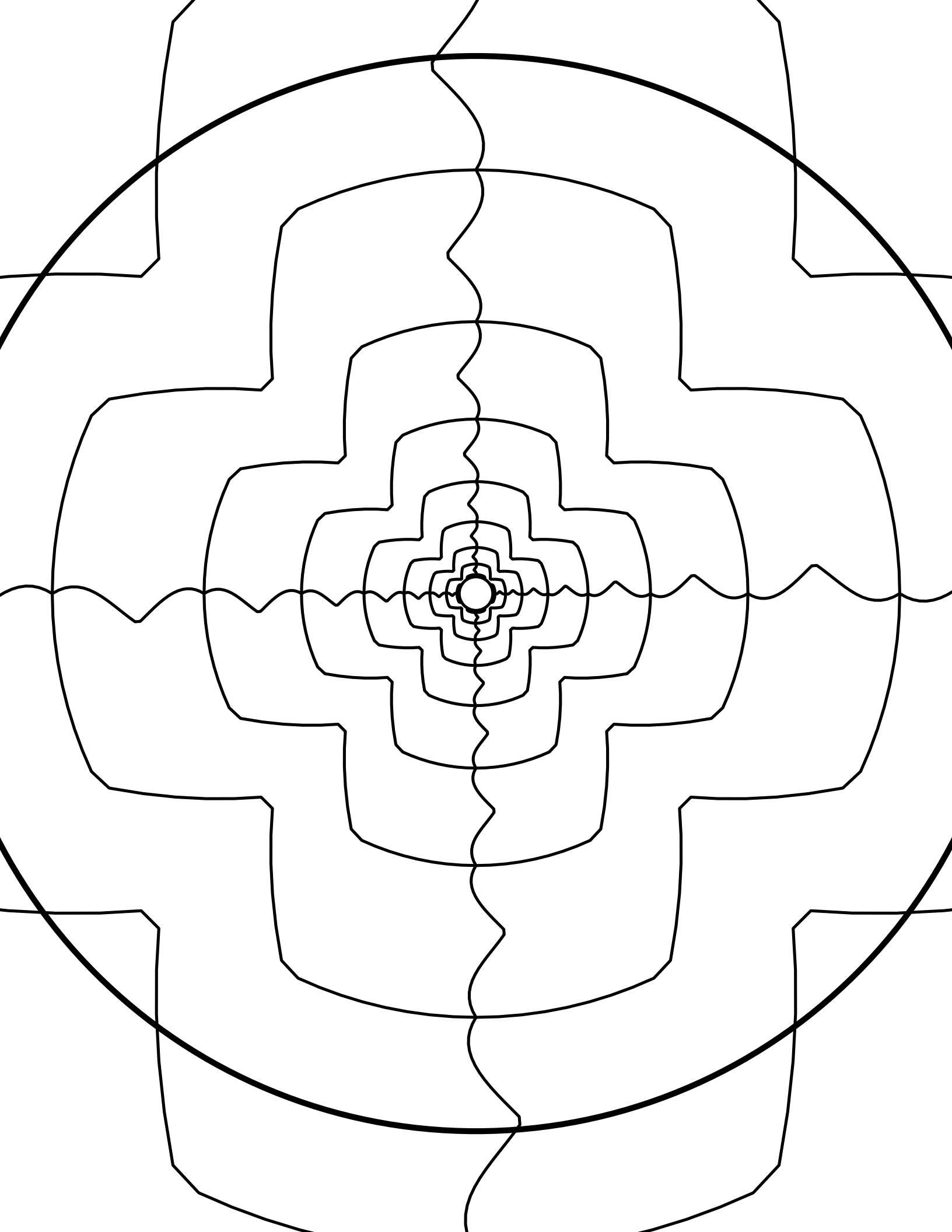
Relax, color from the center outward, and watch the math unfold.

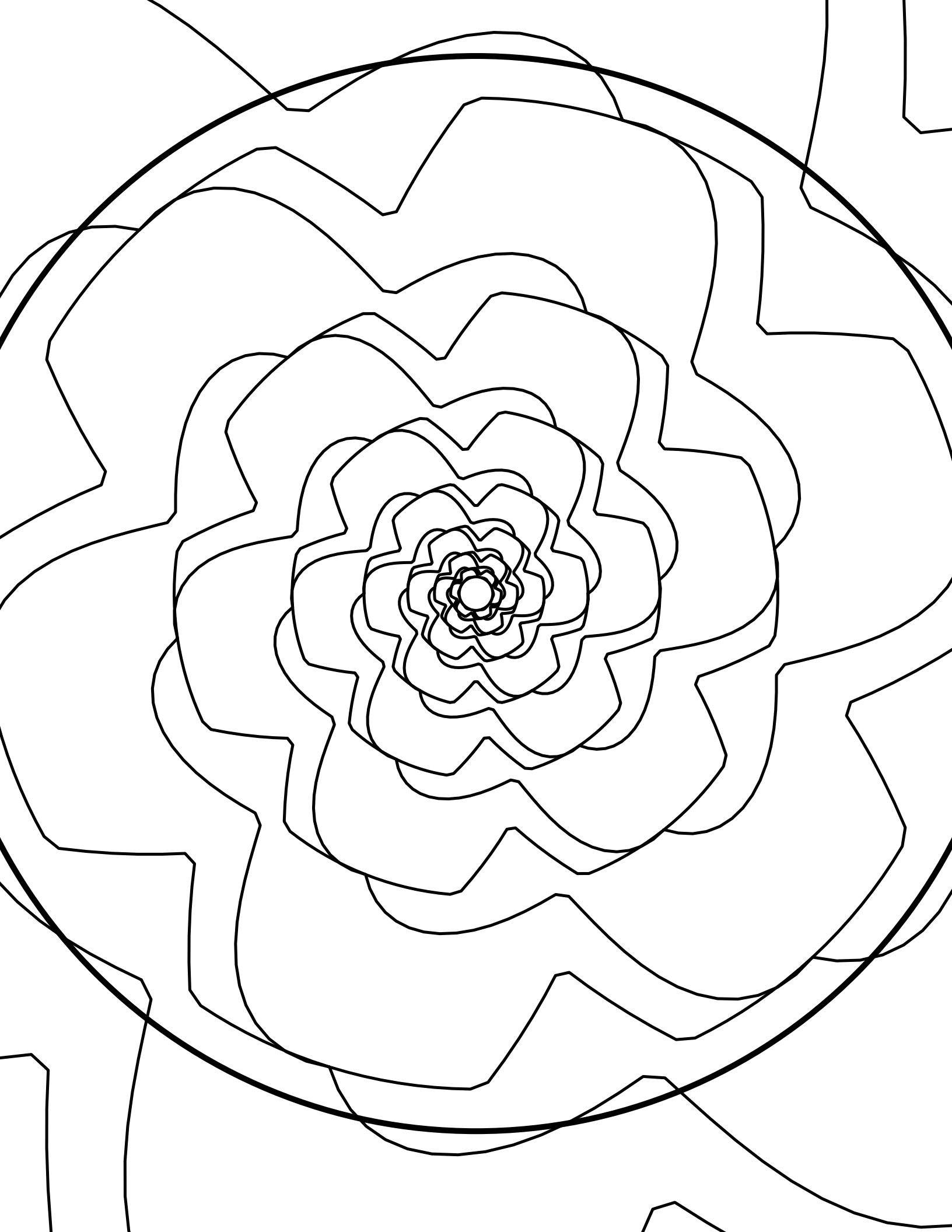
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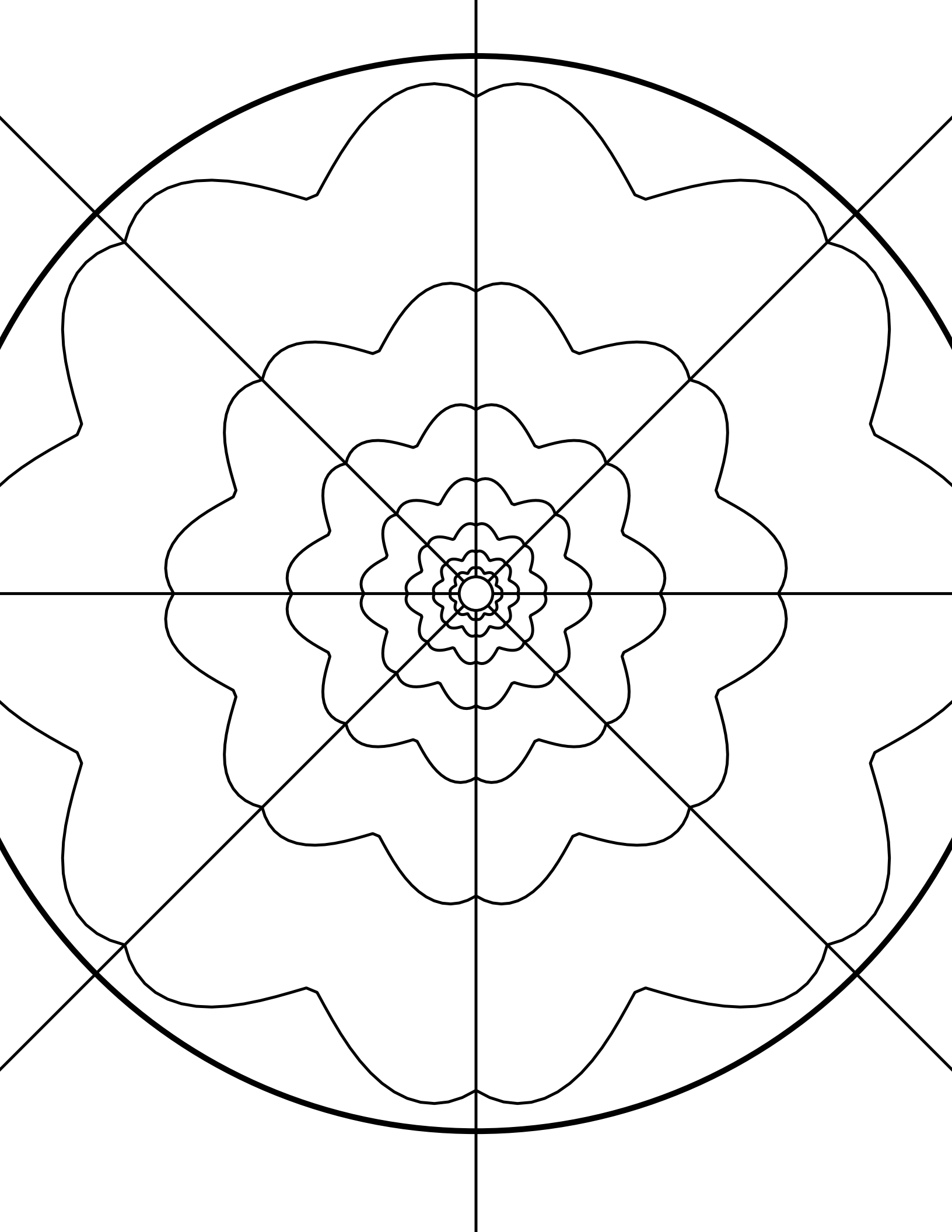


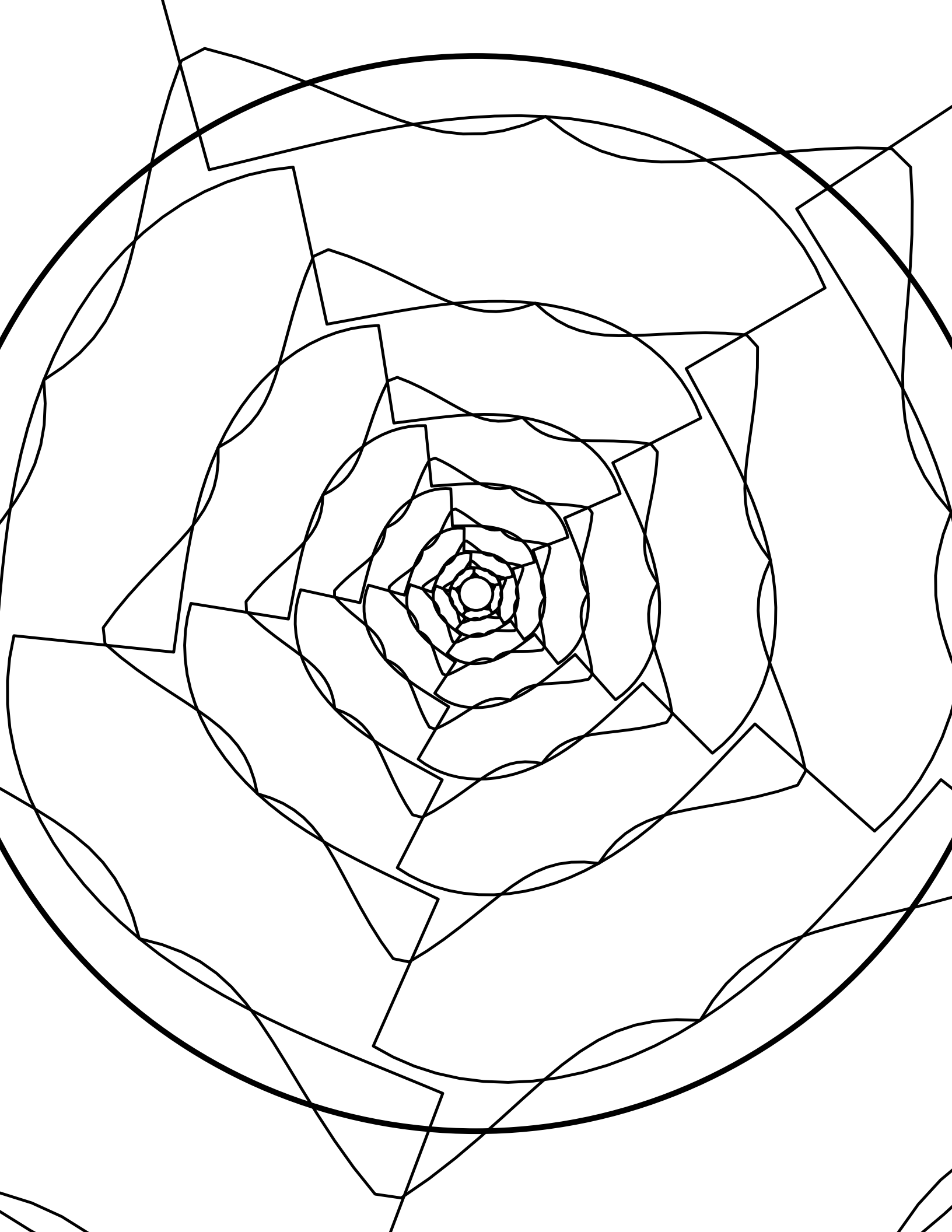


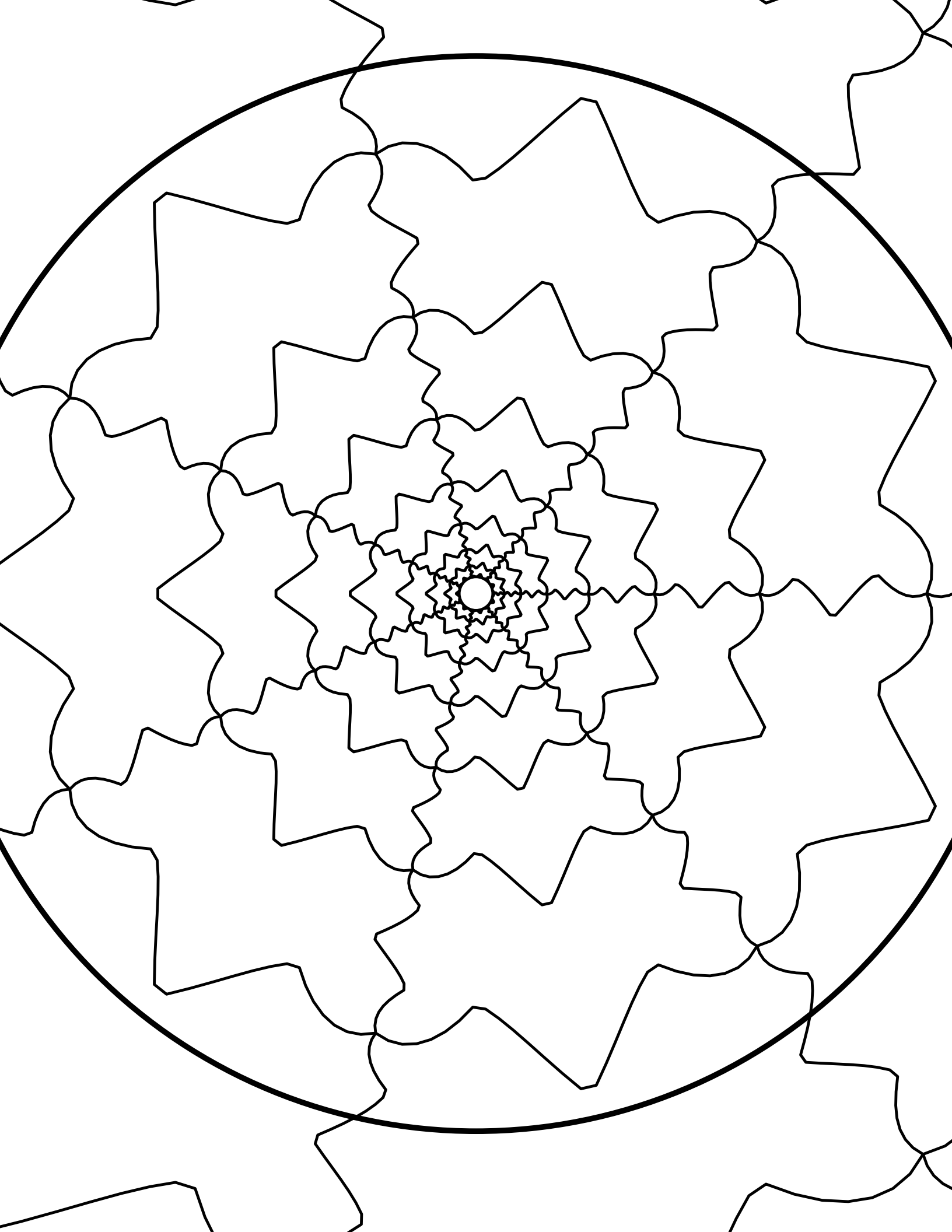


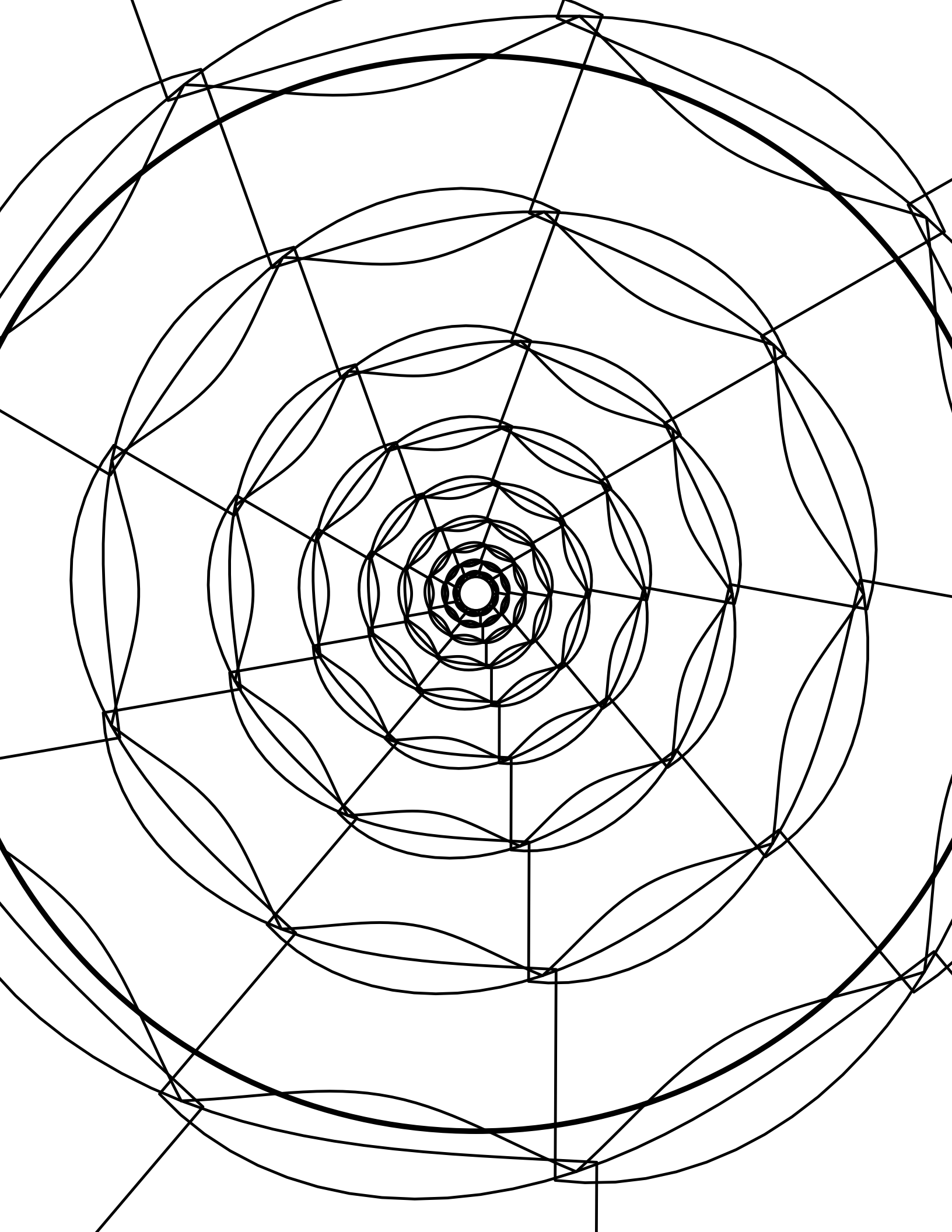


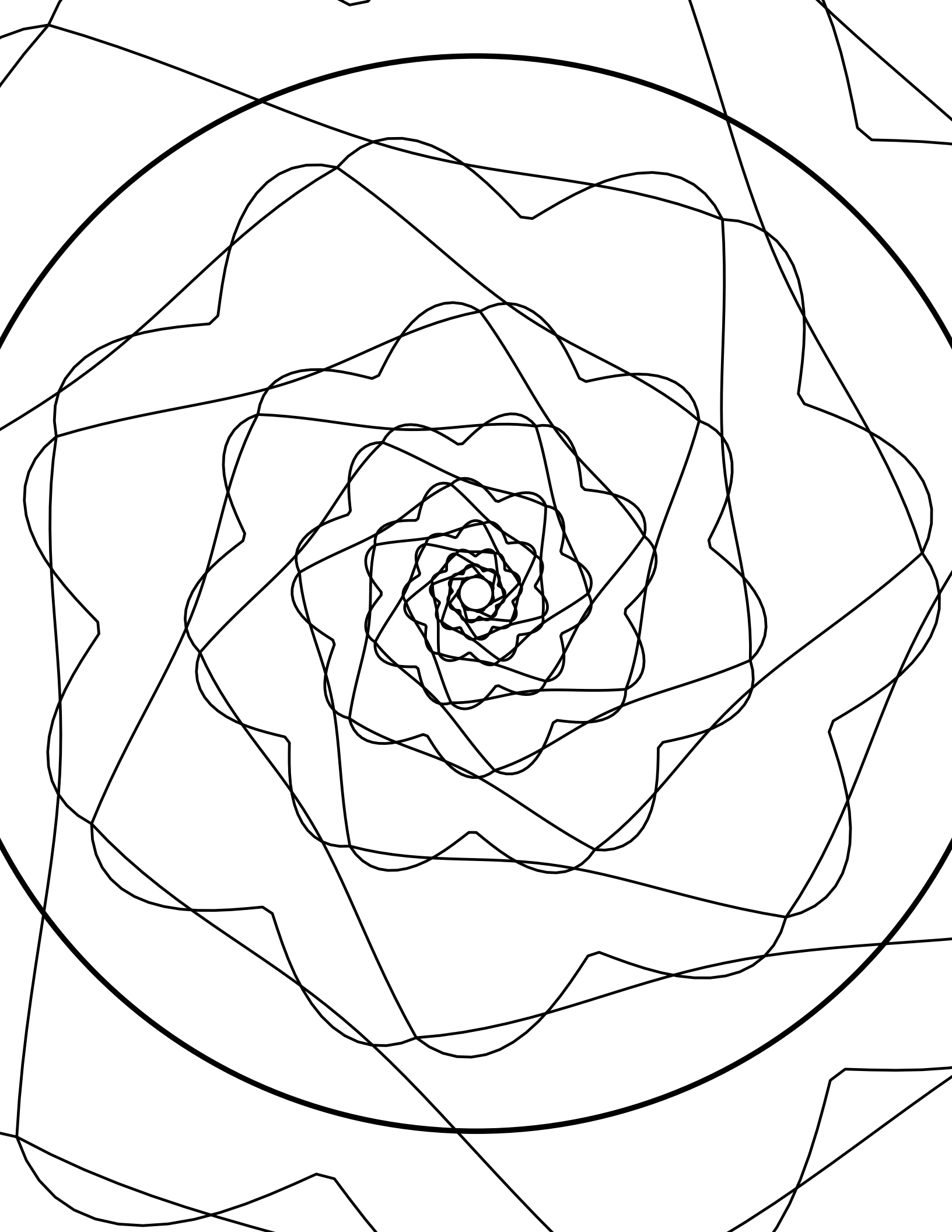


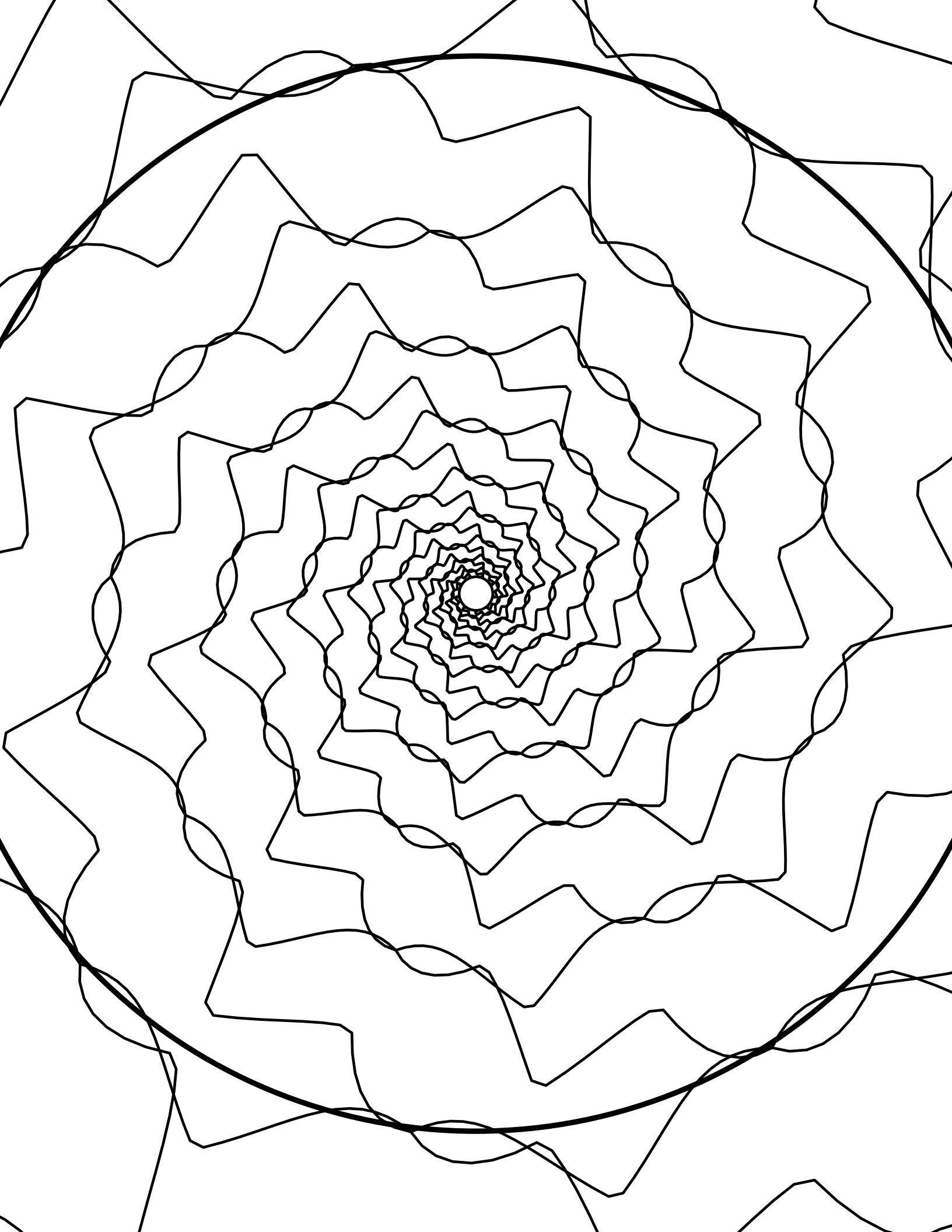


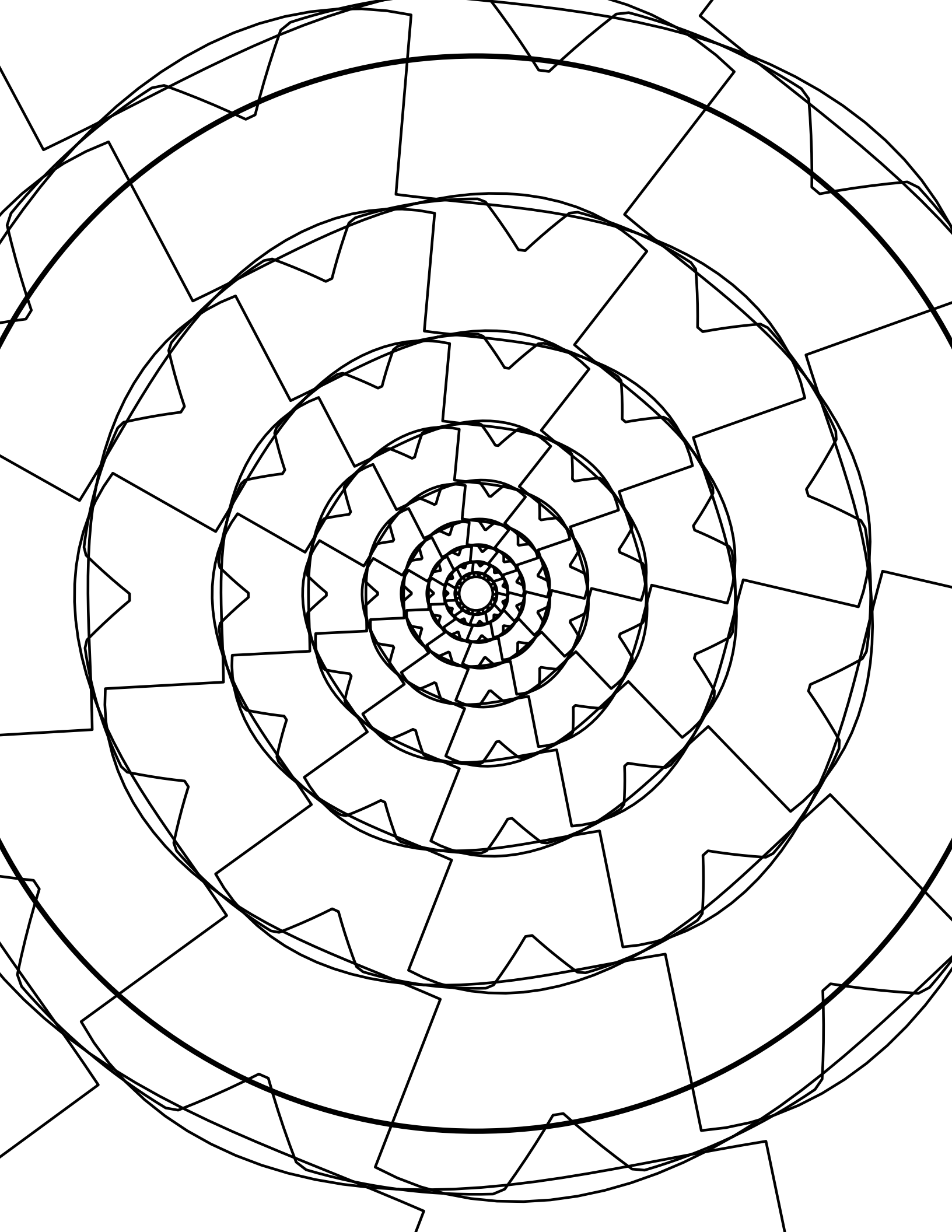


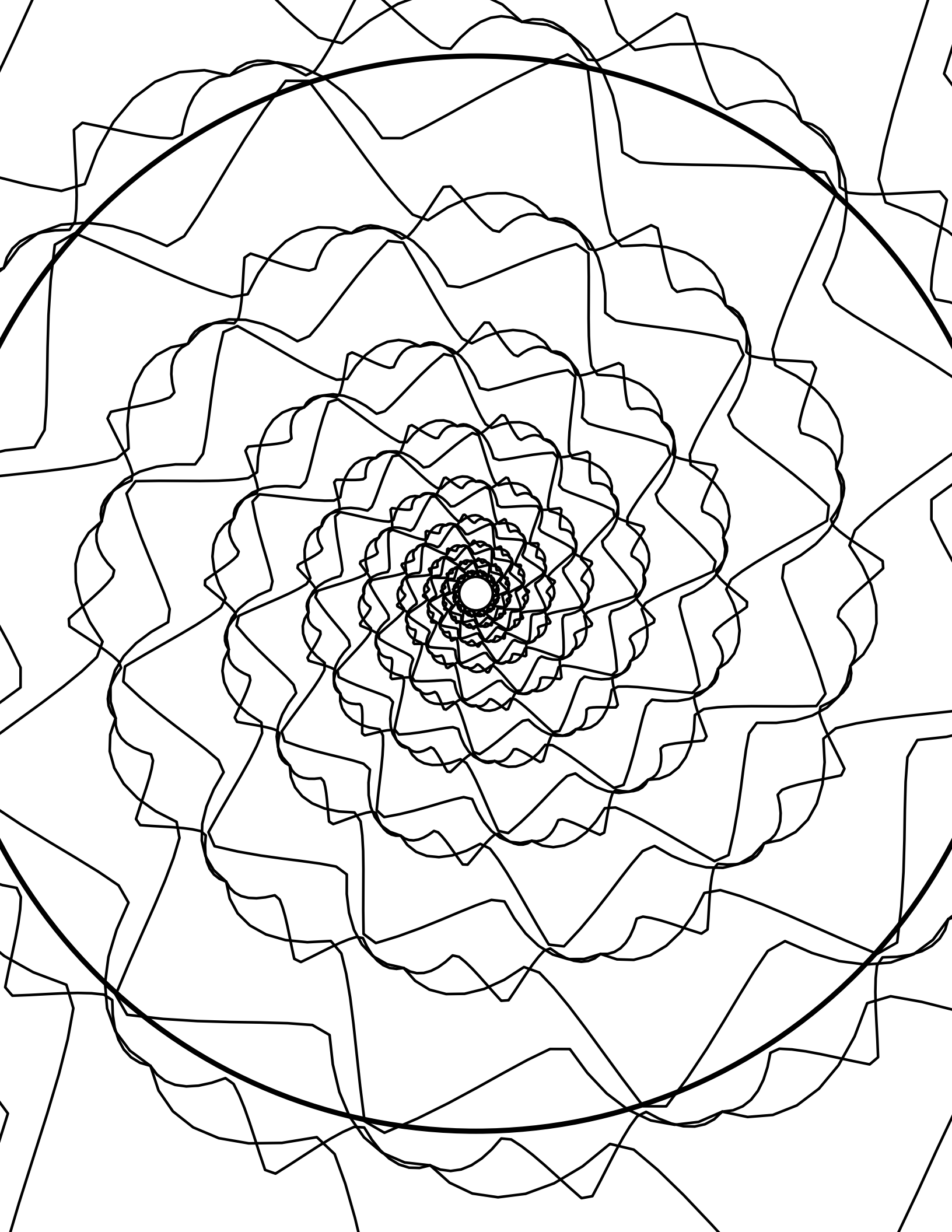


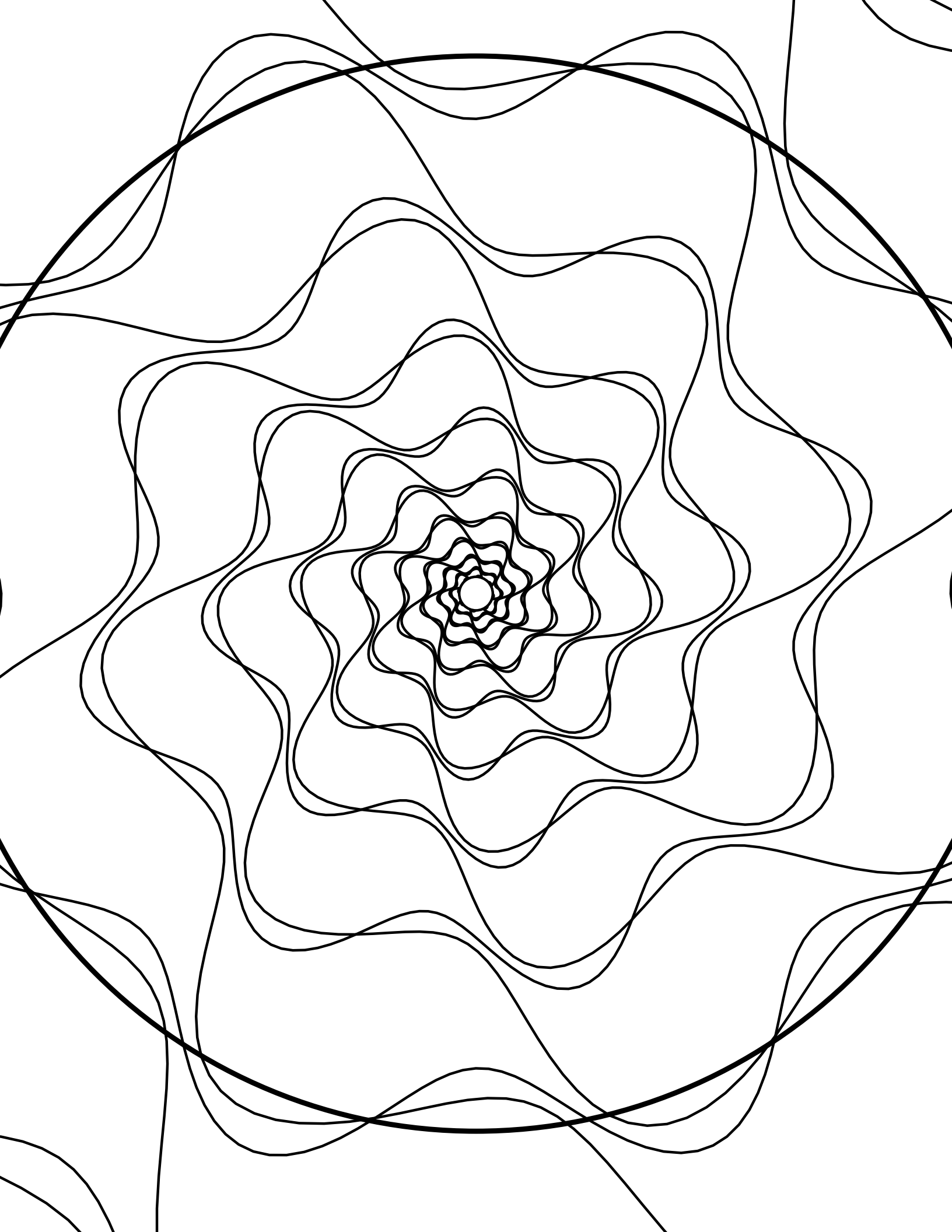


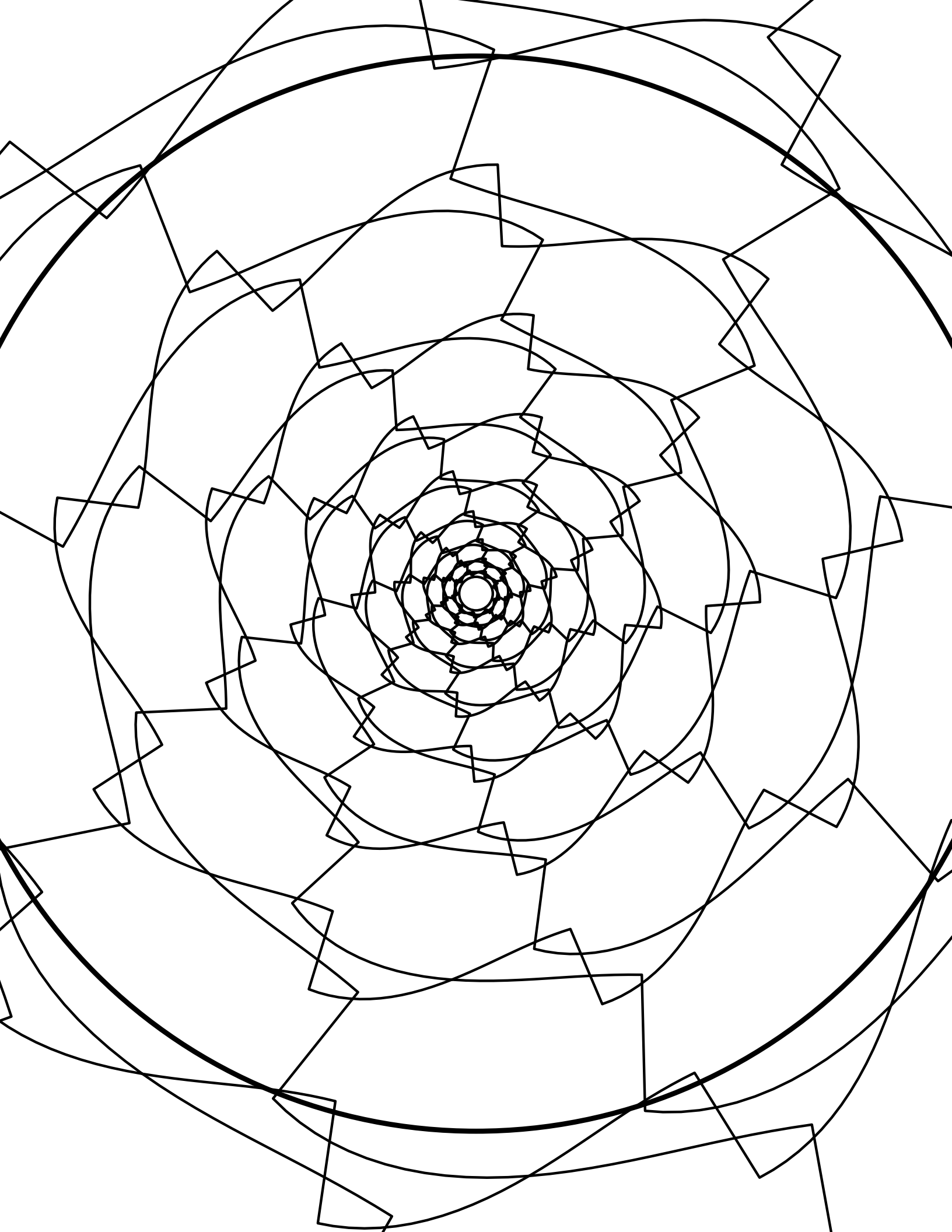




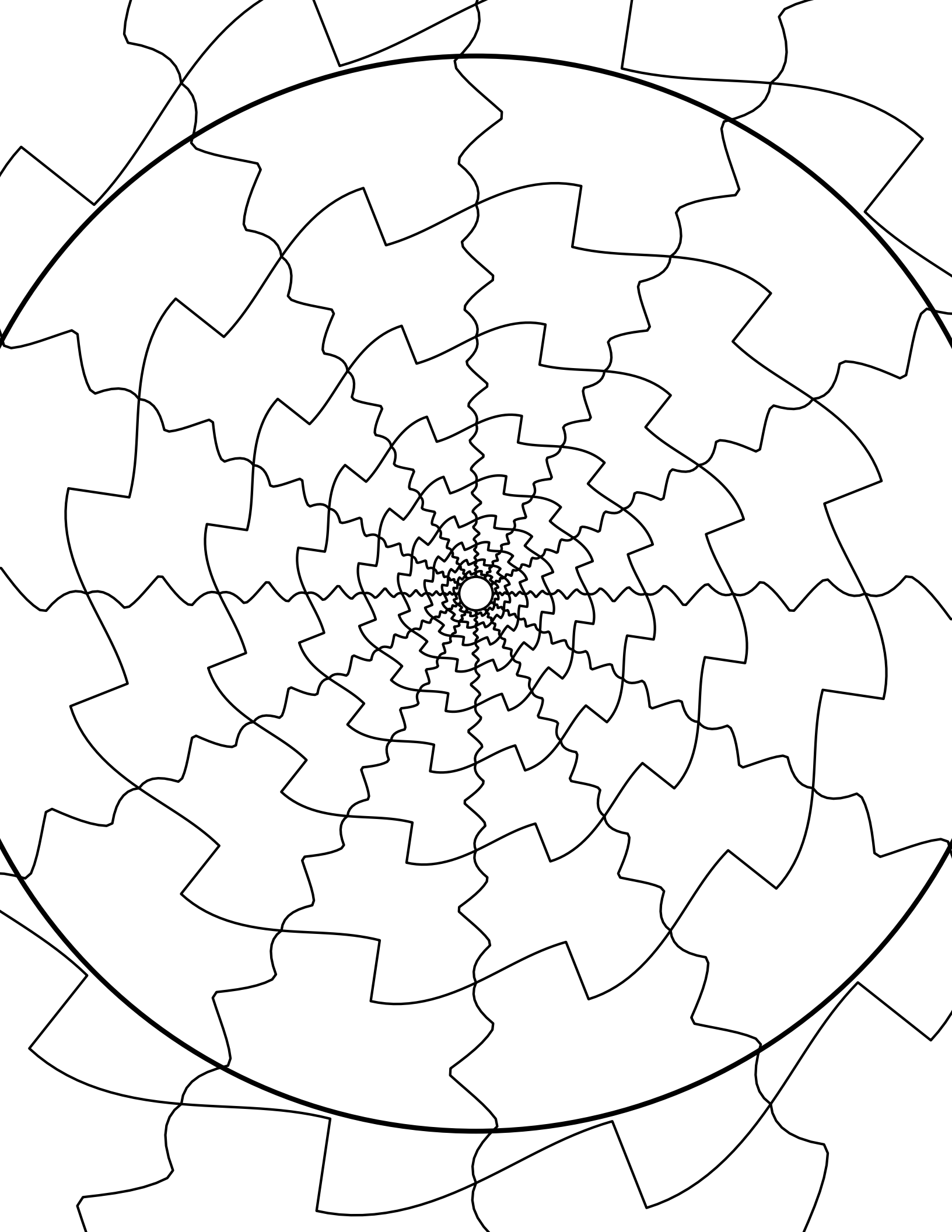


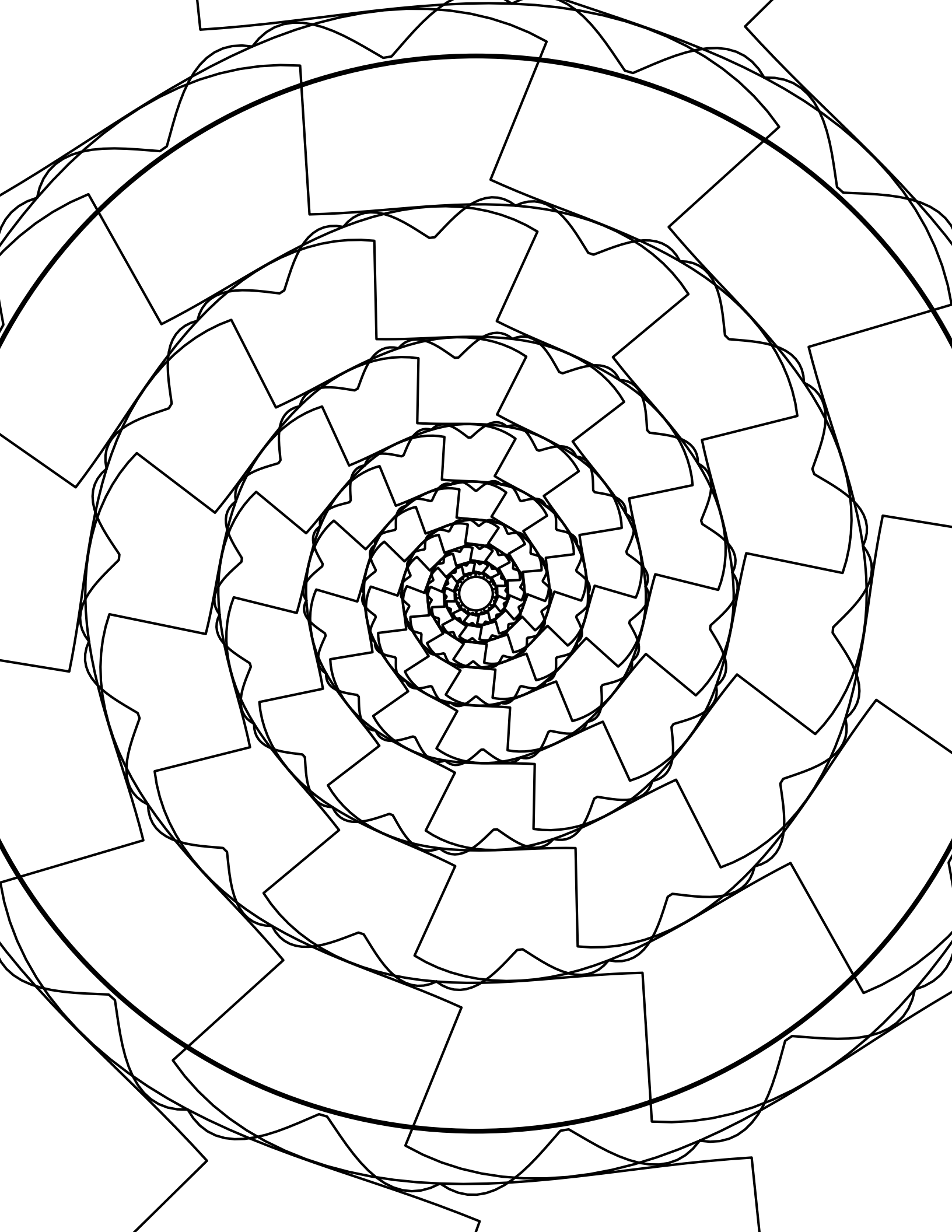


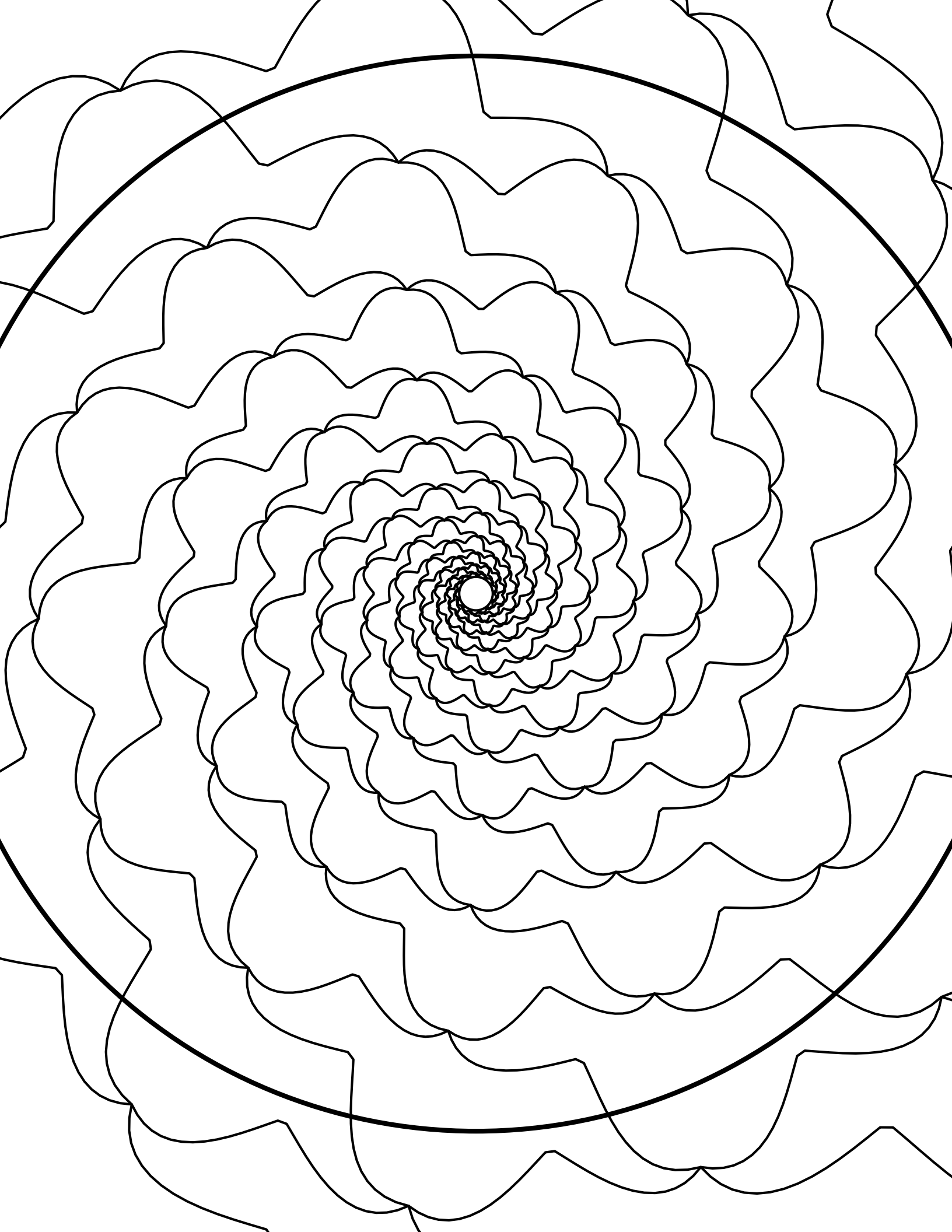


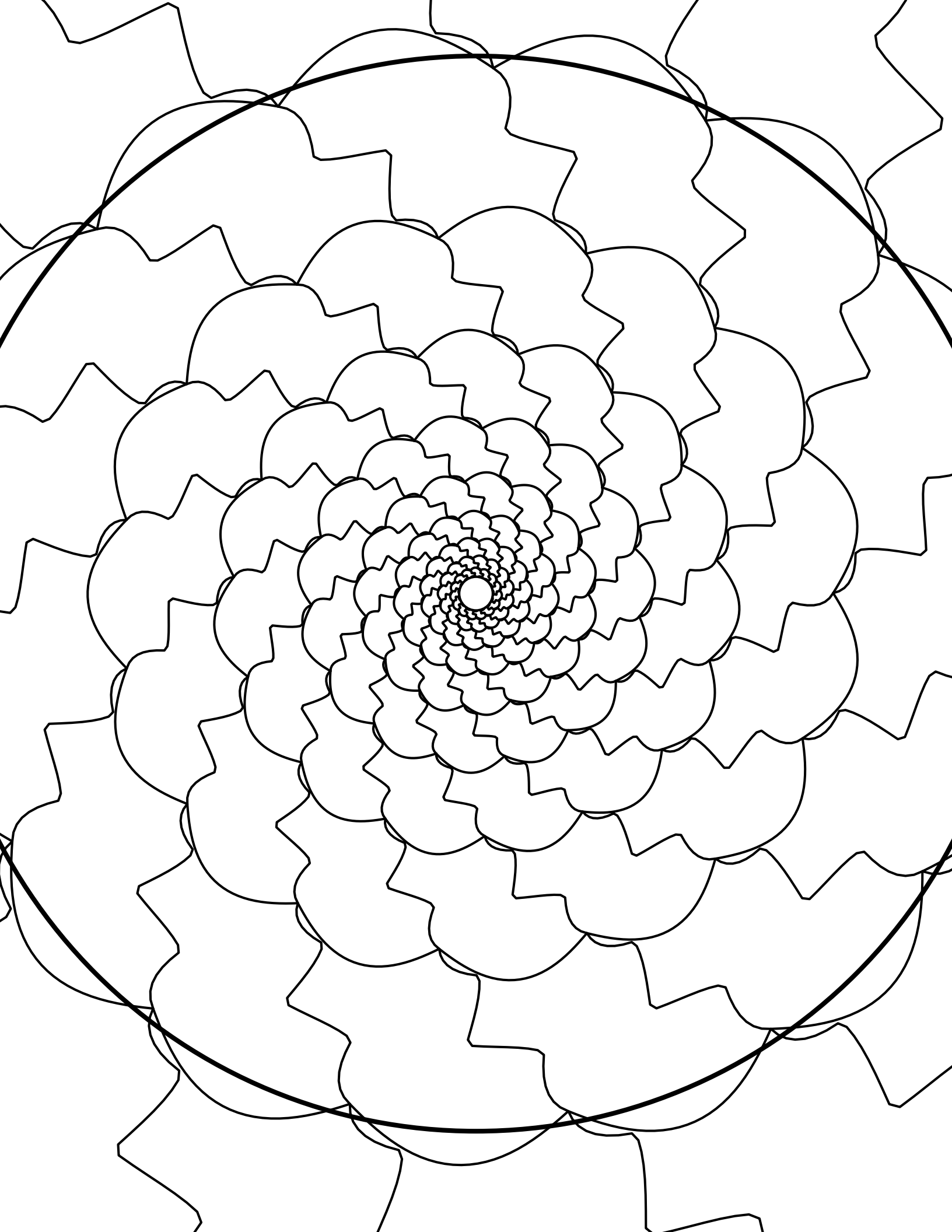


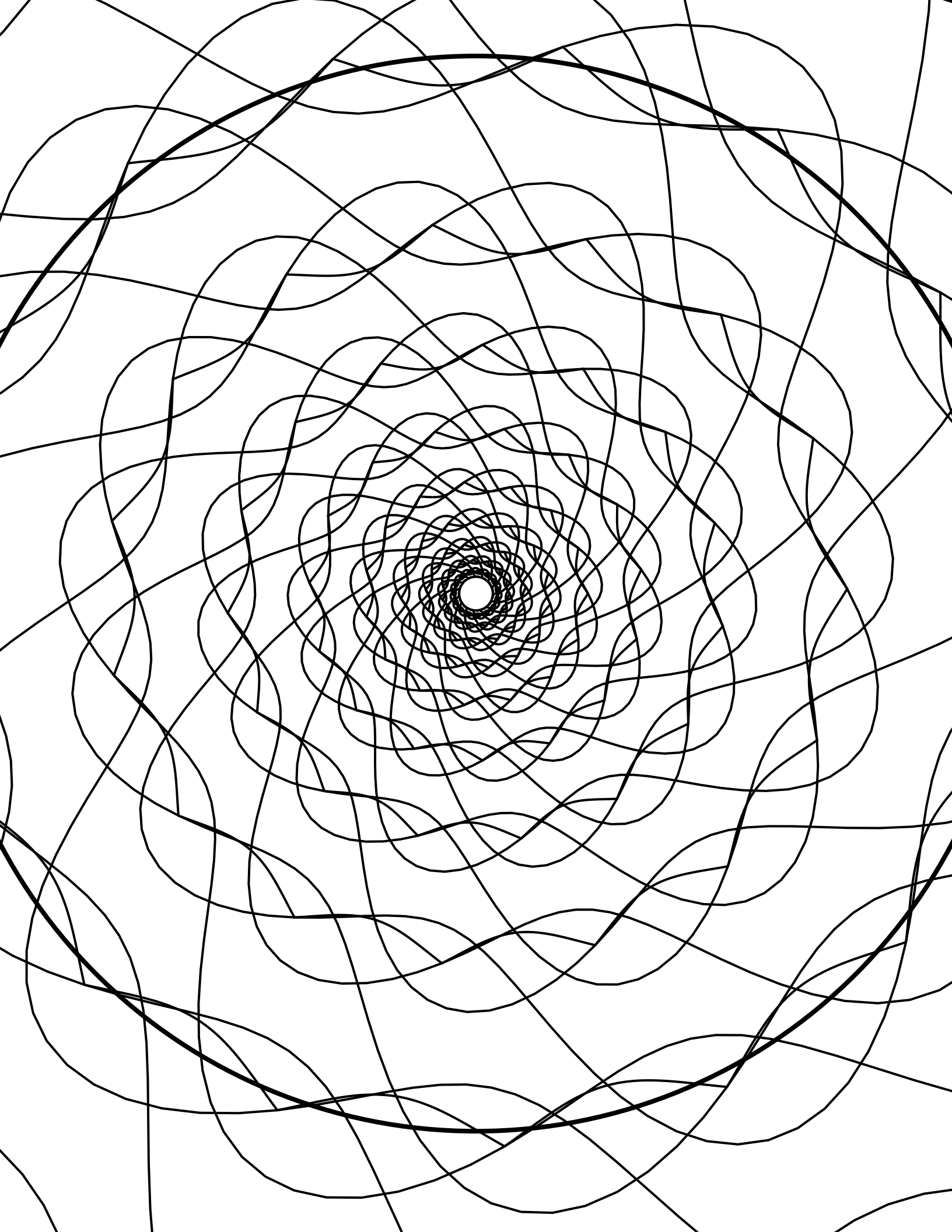


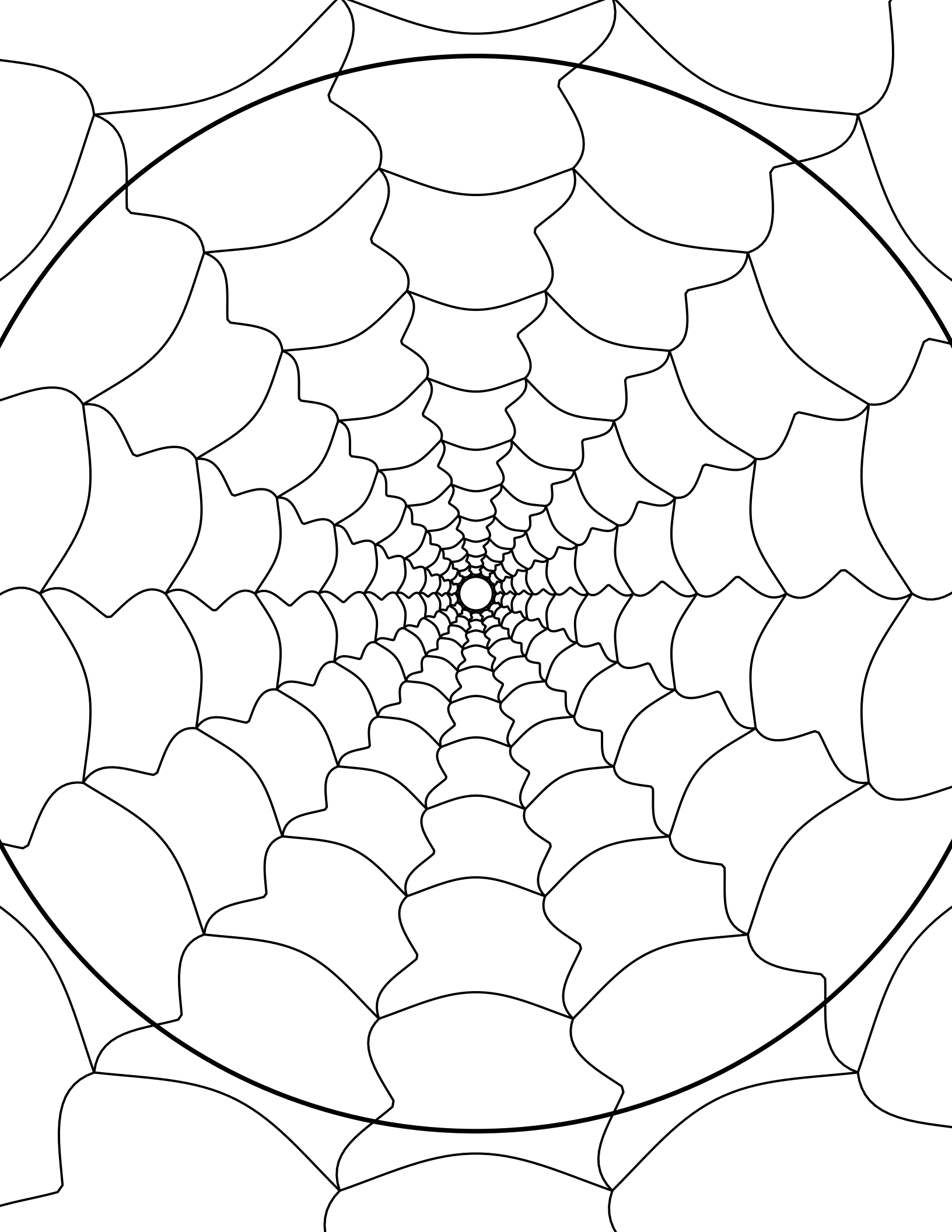


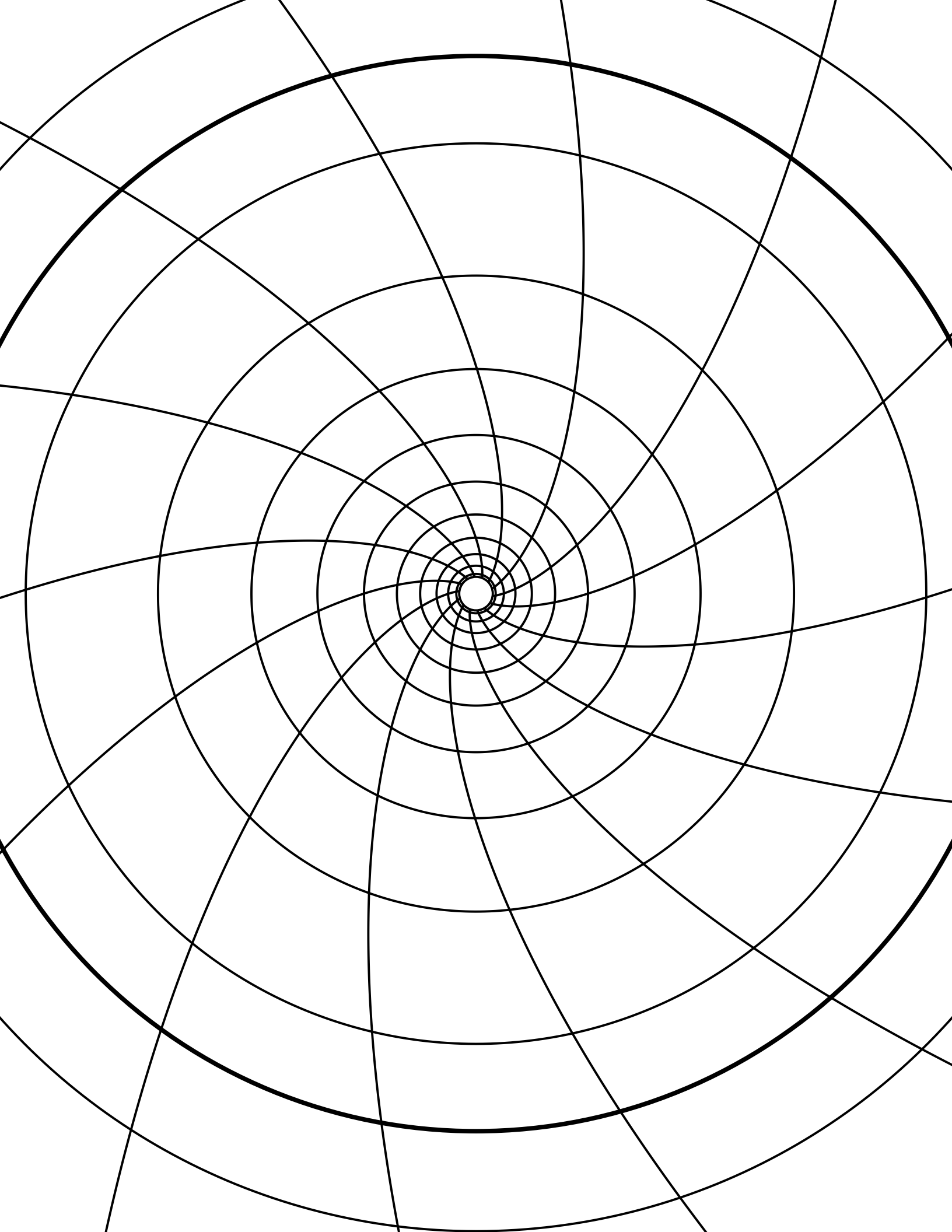


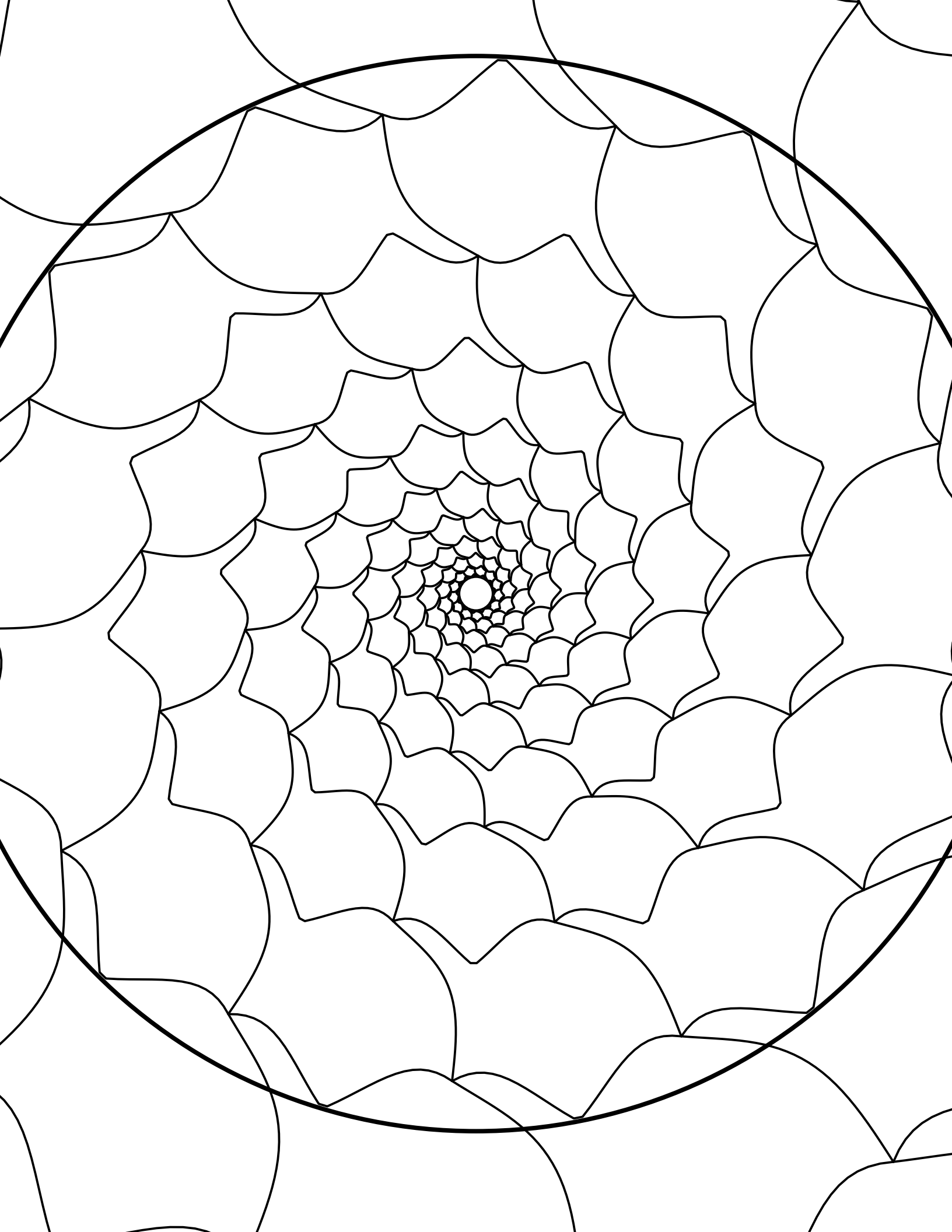


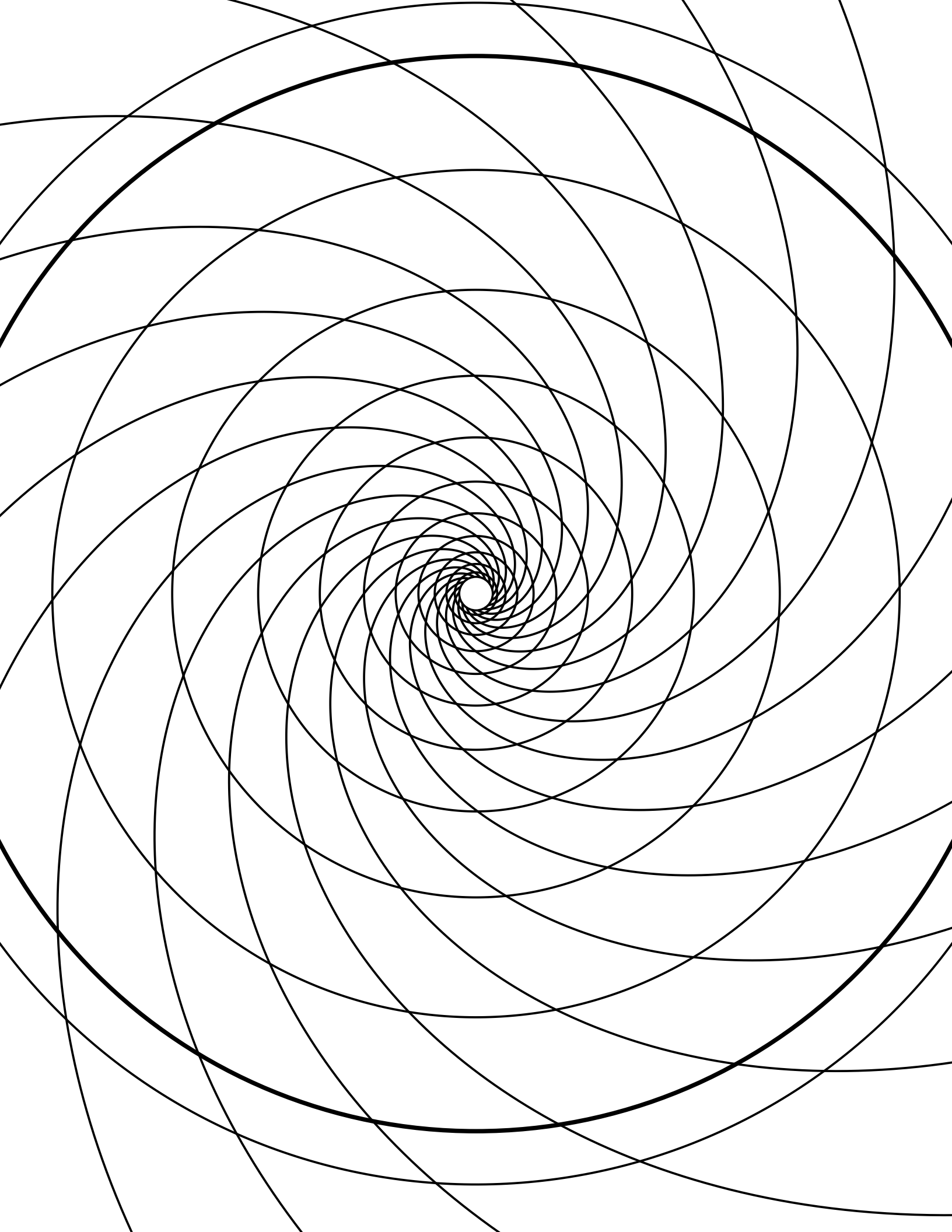


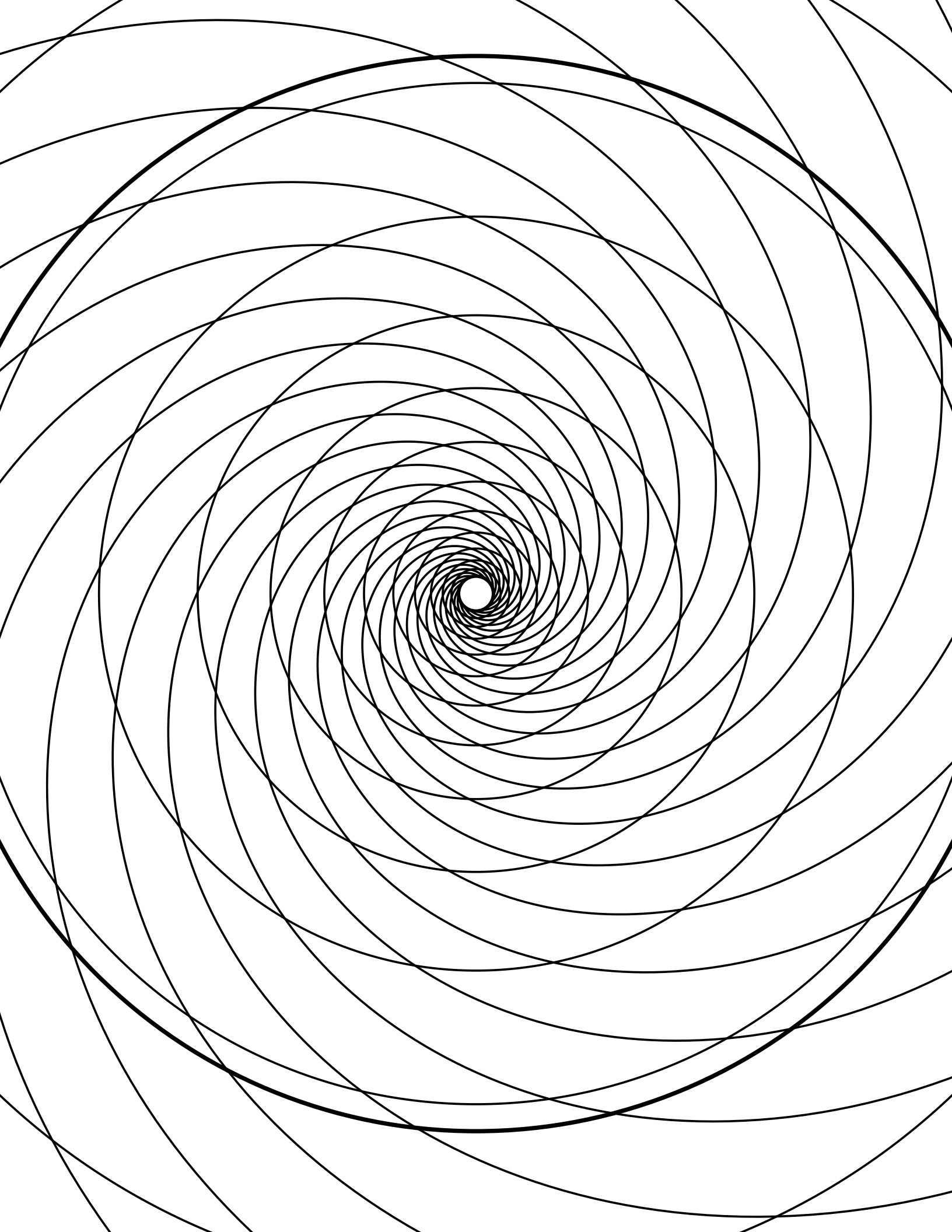


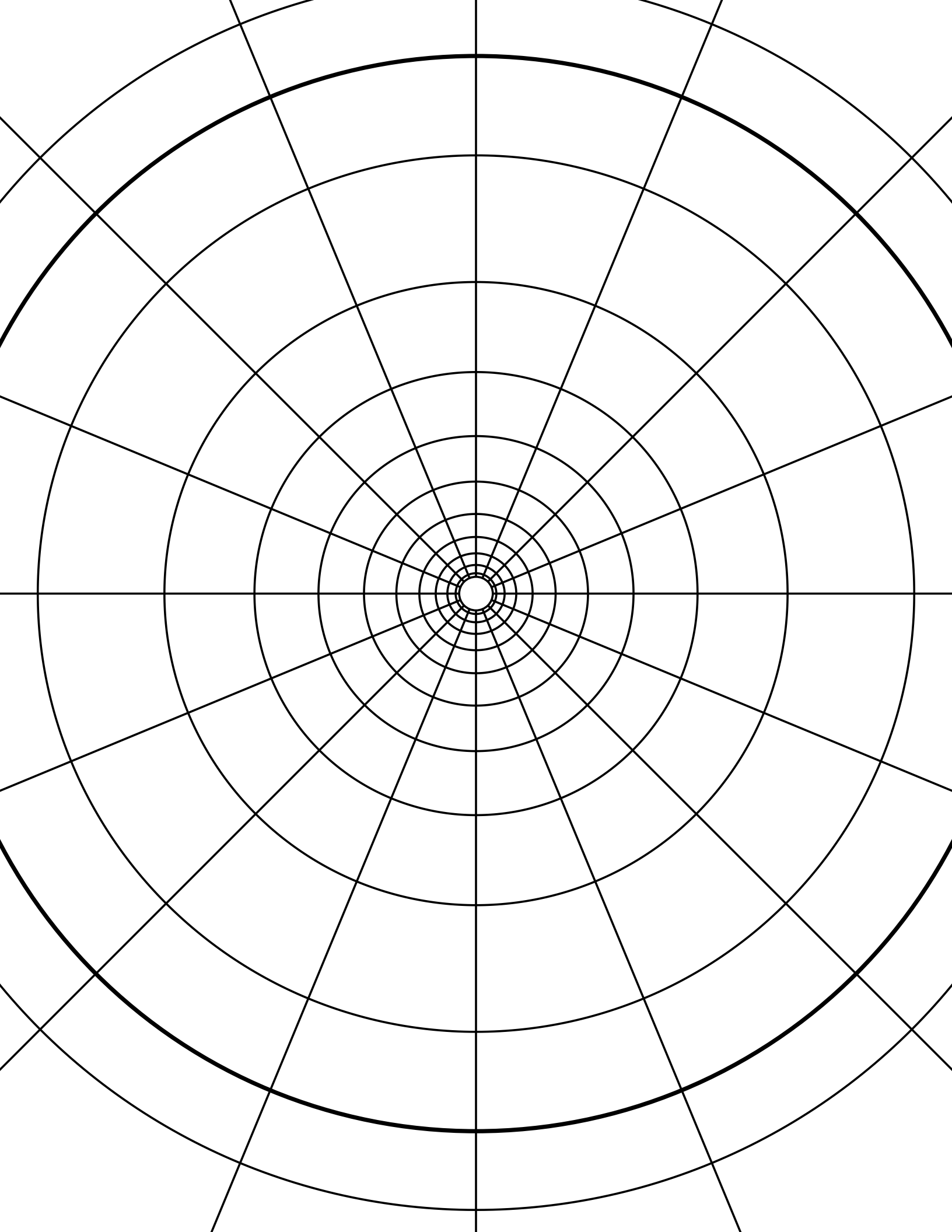


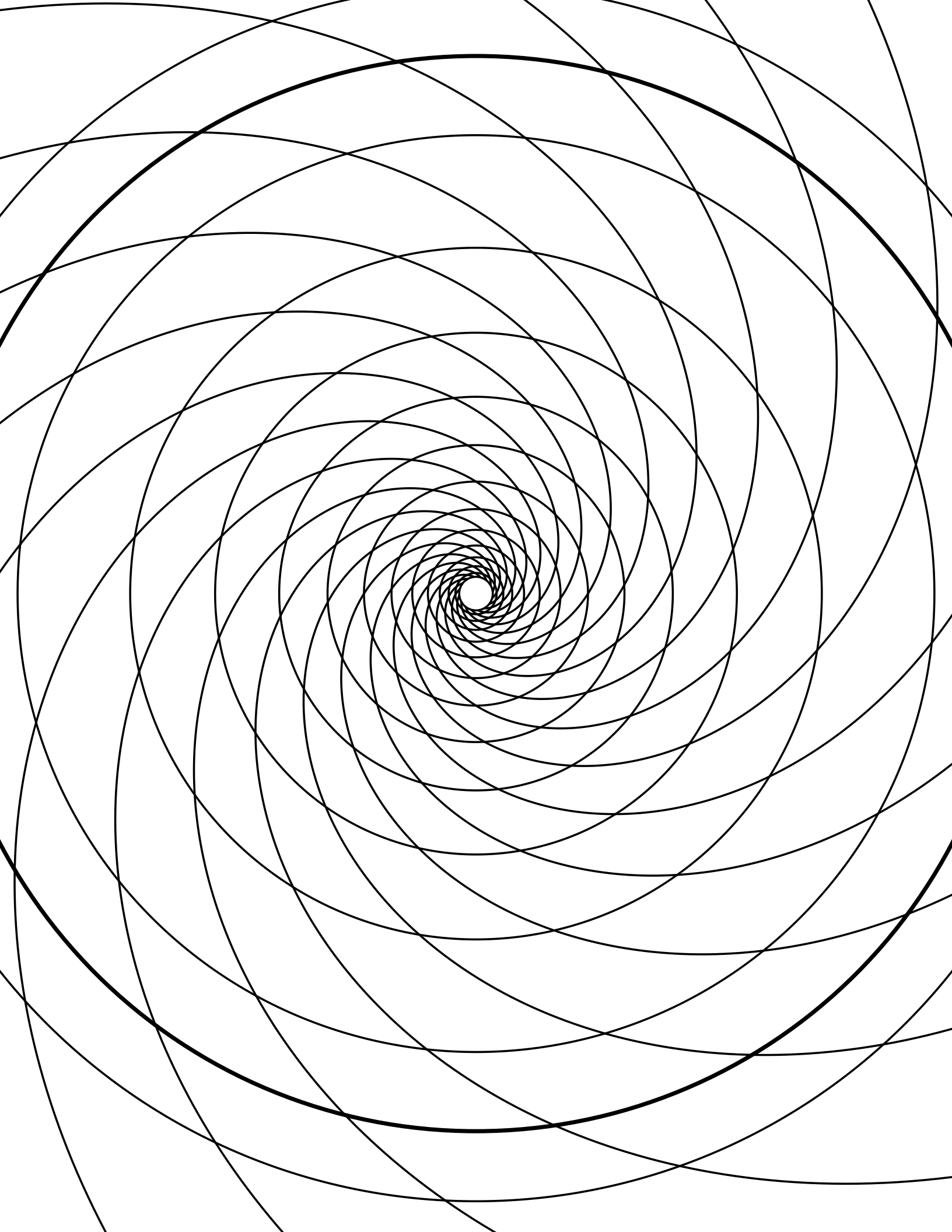


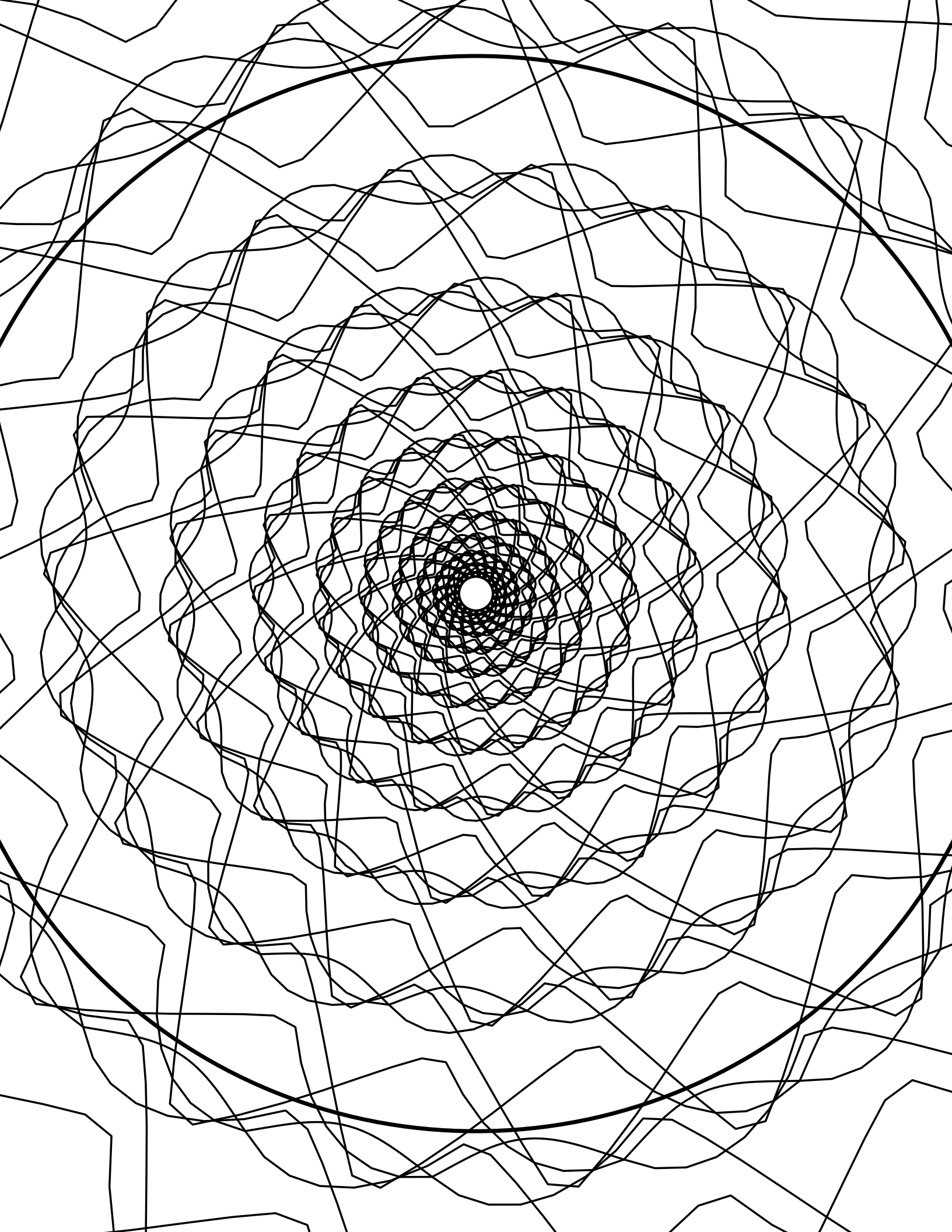


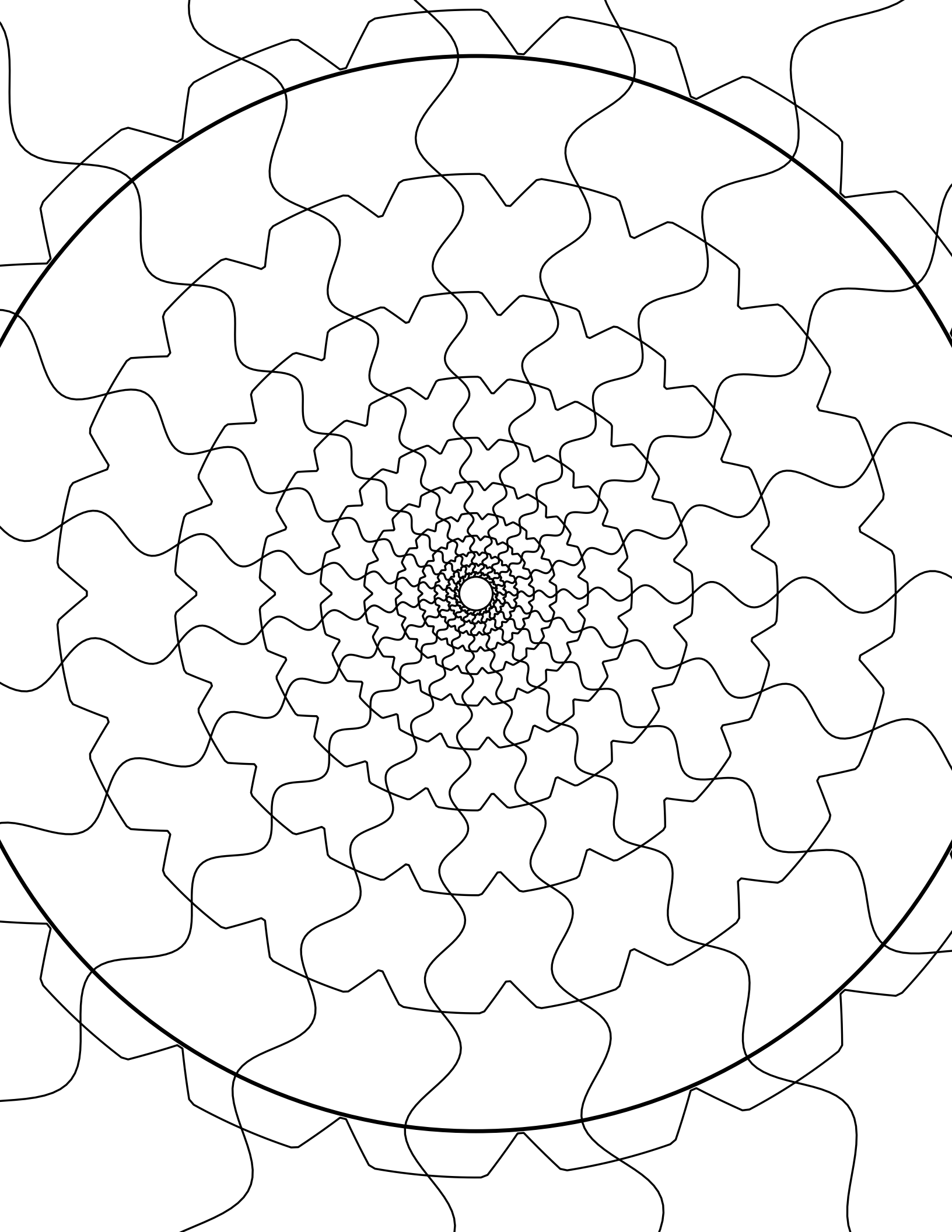


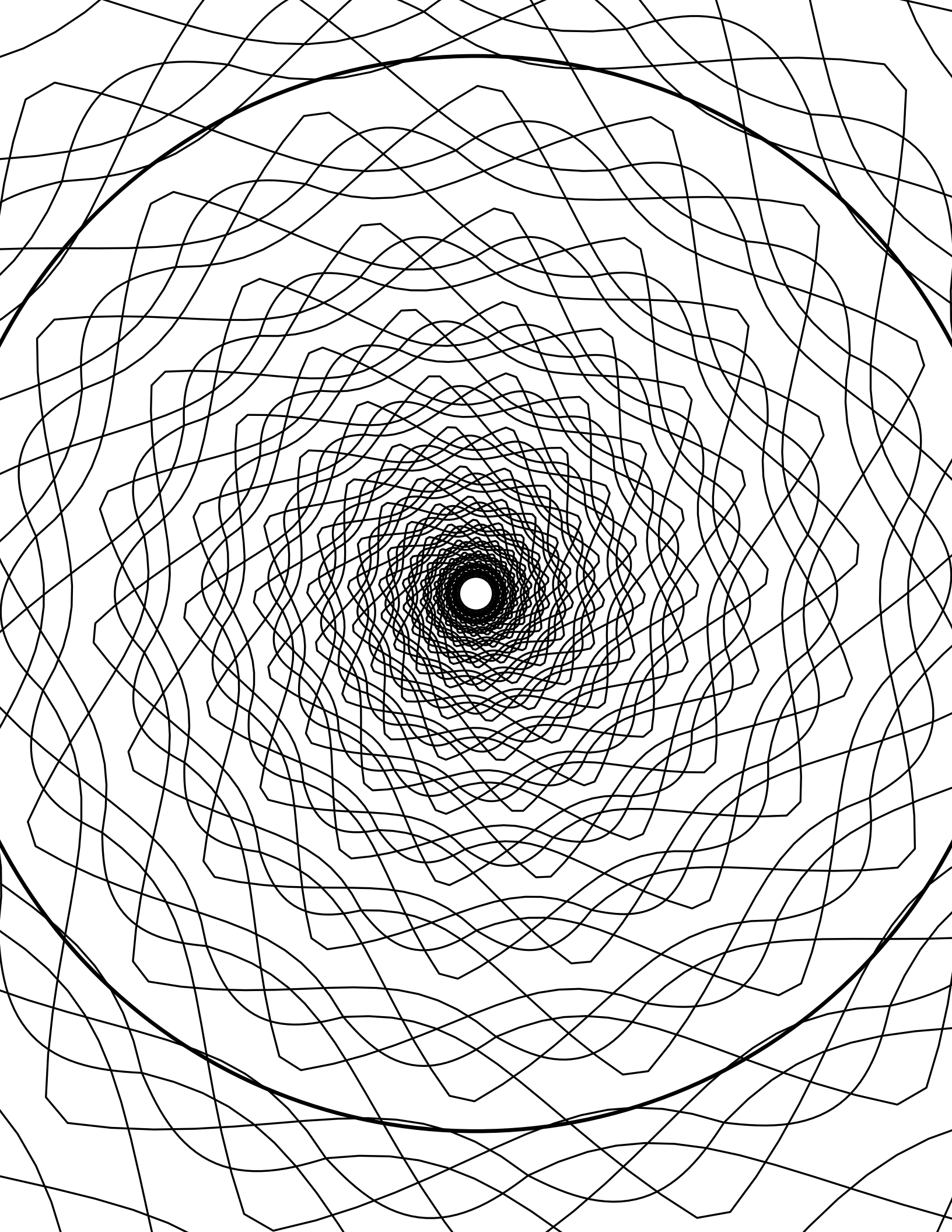


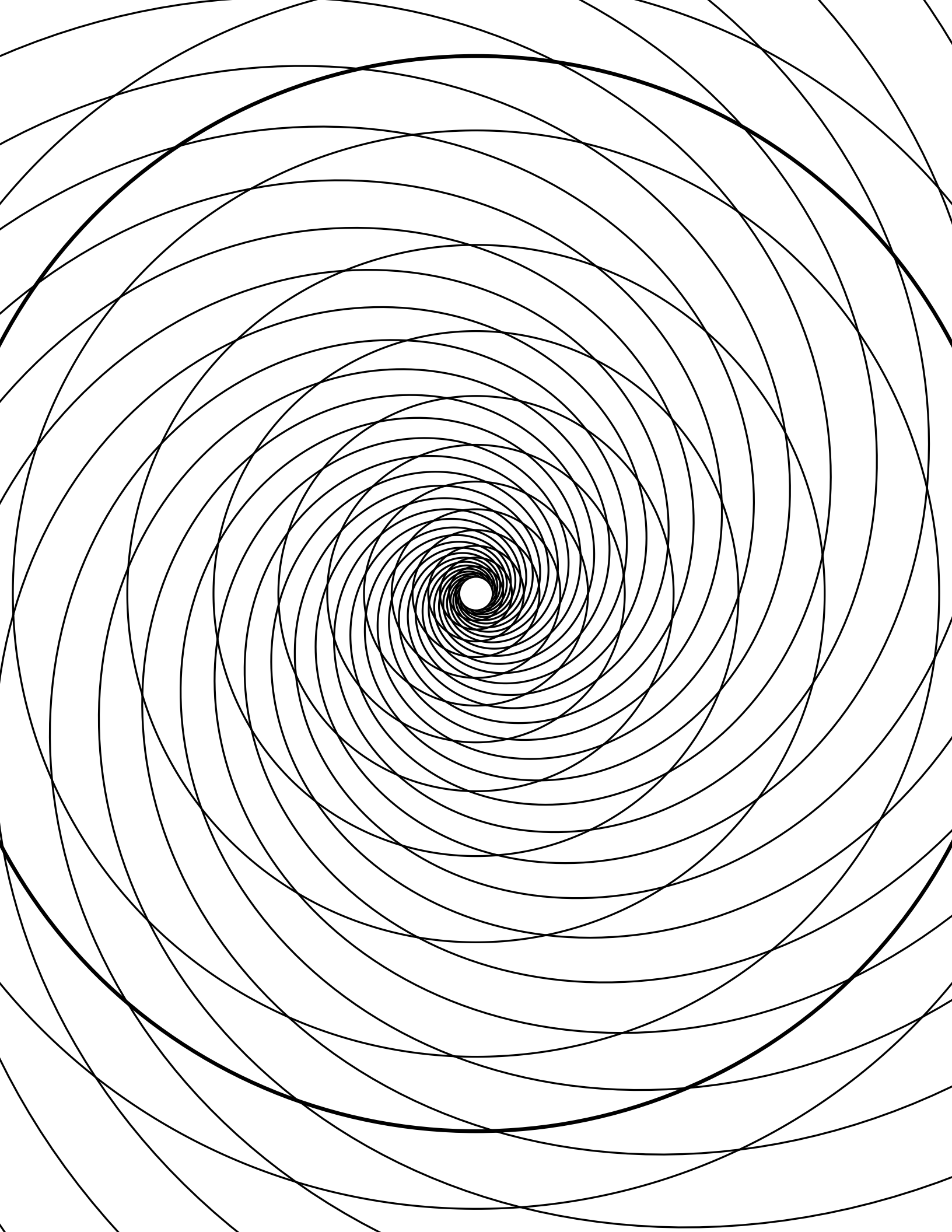


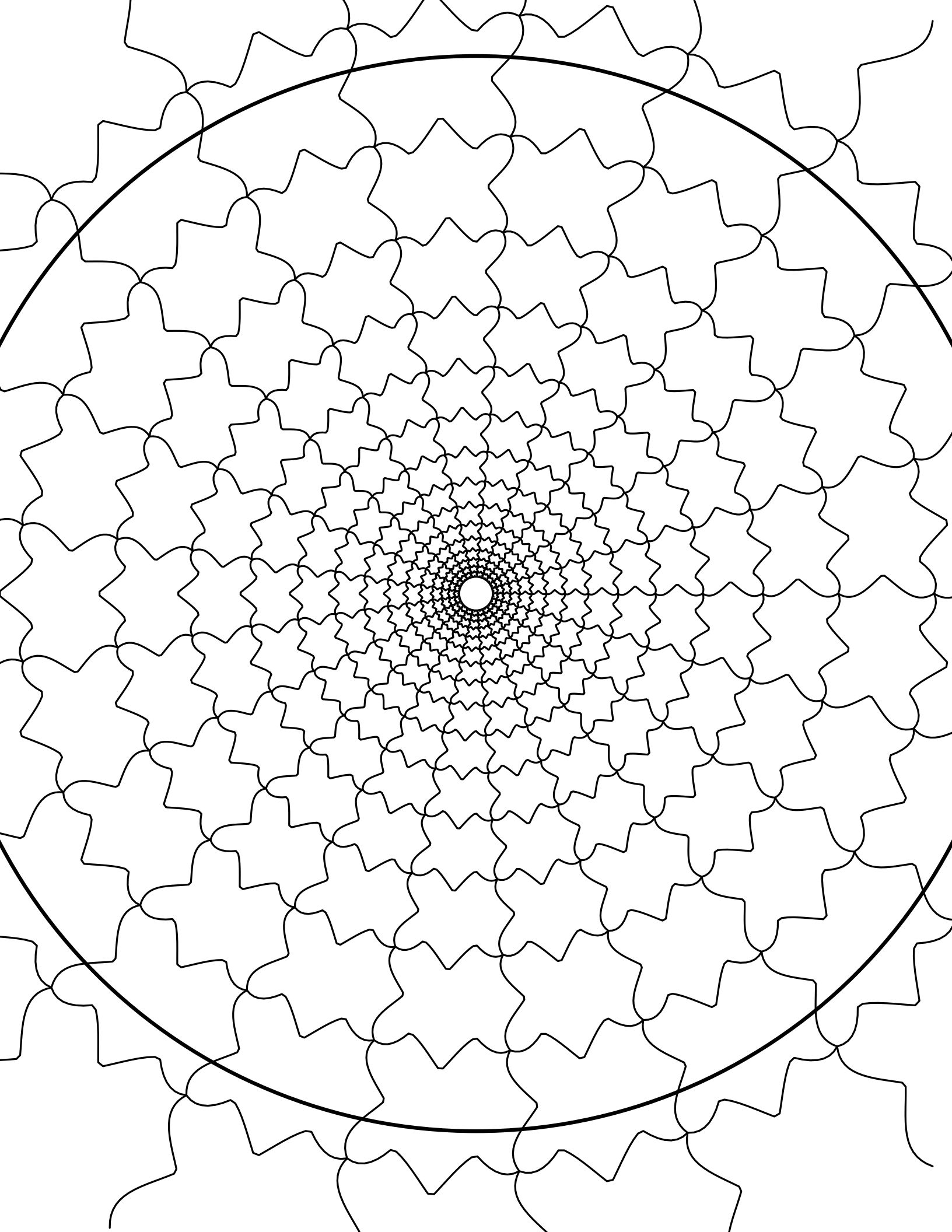


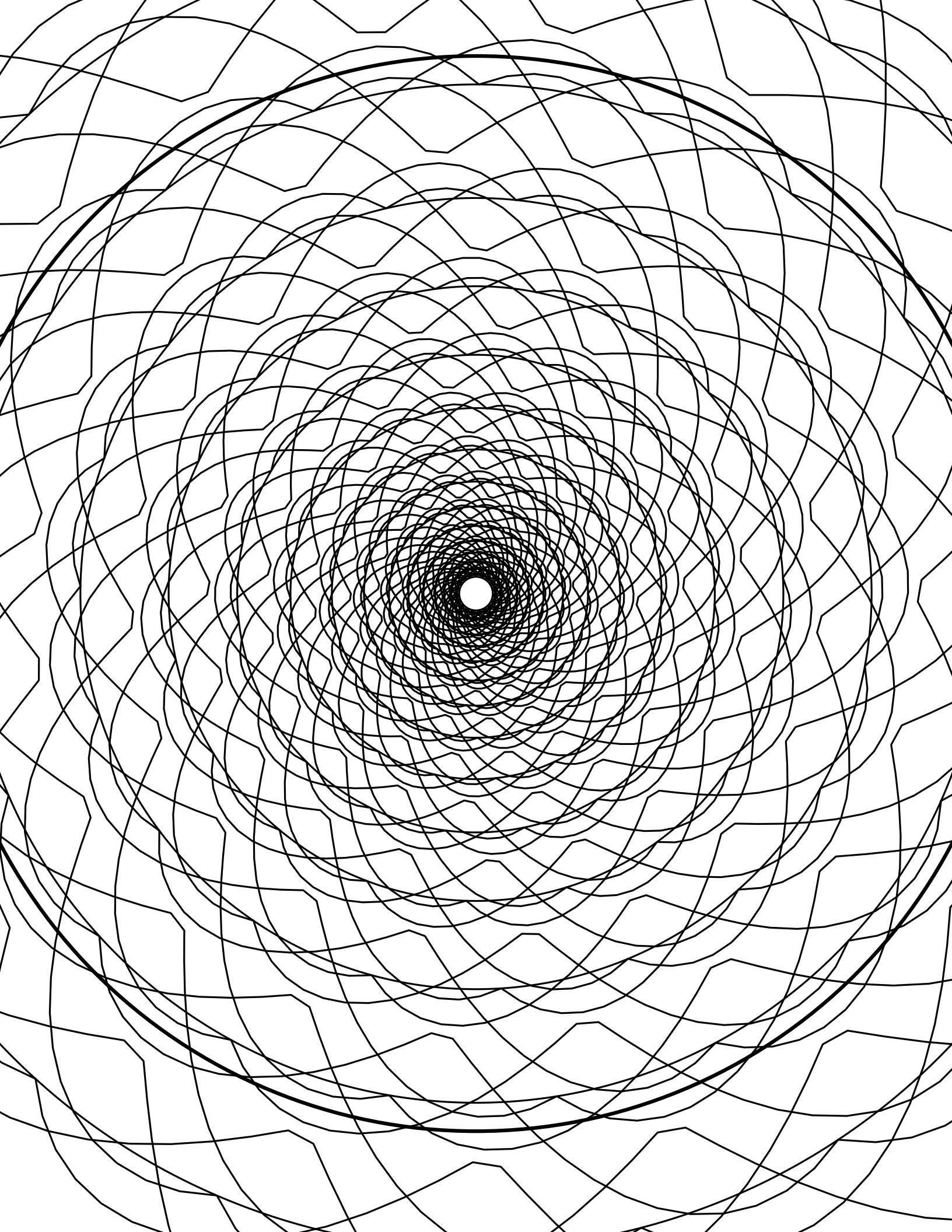


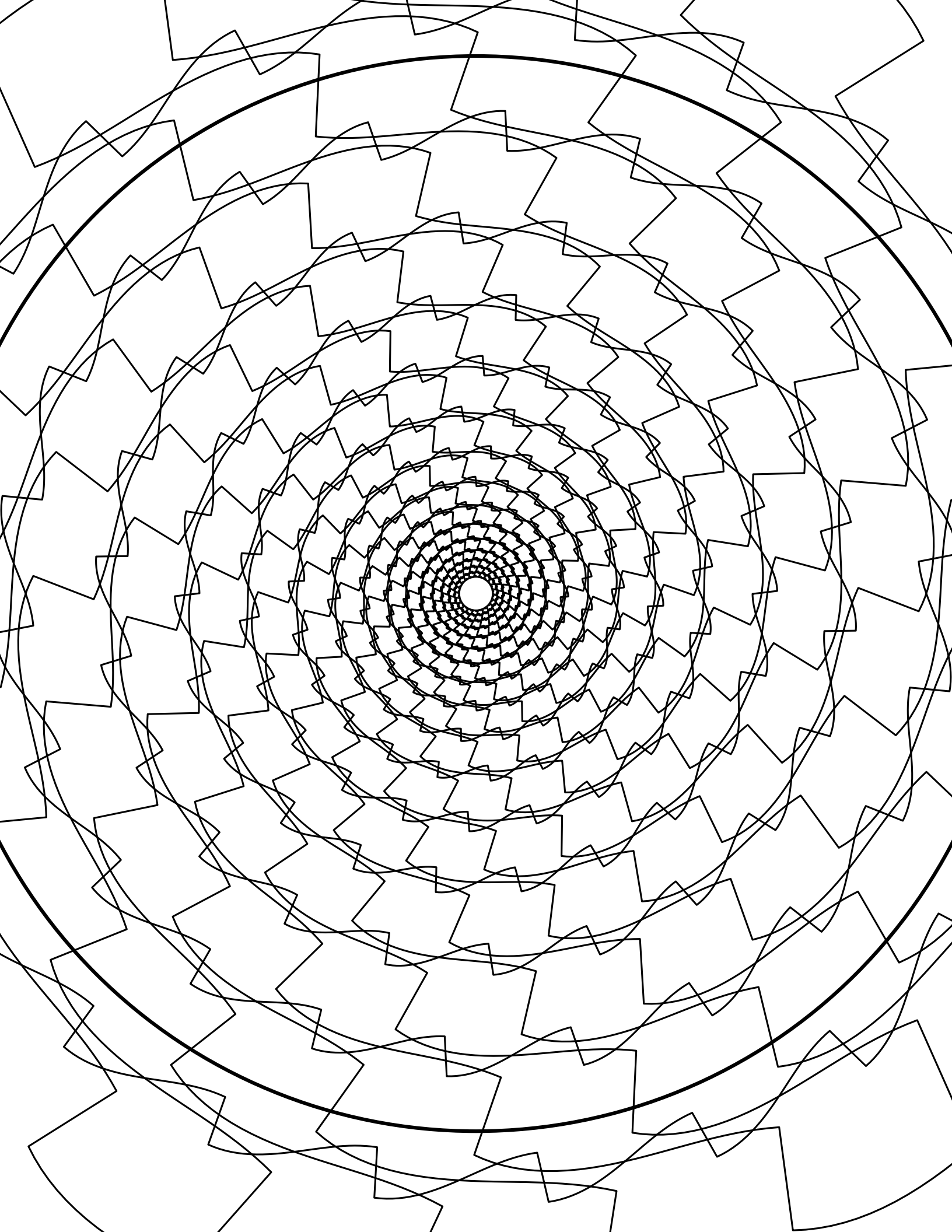


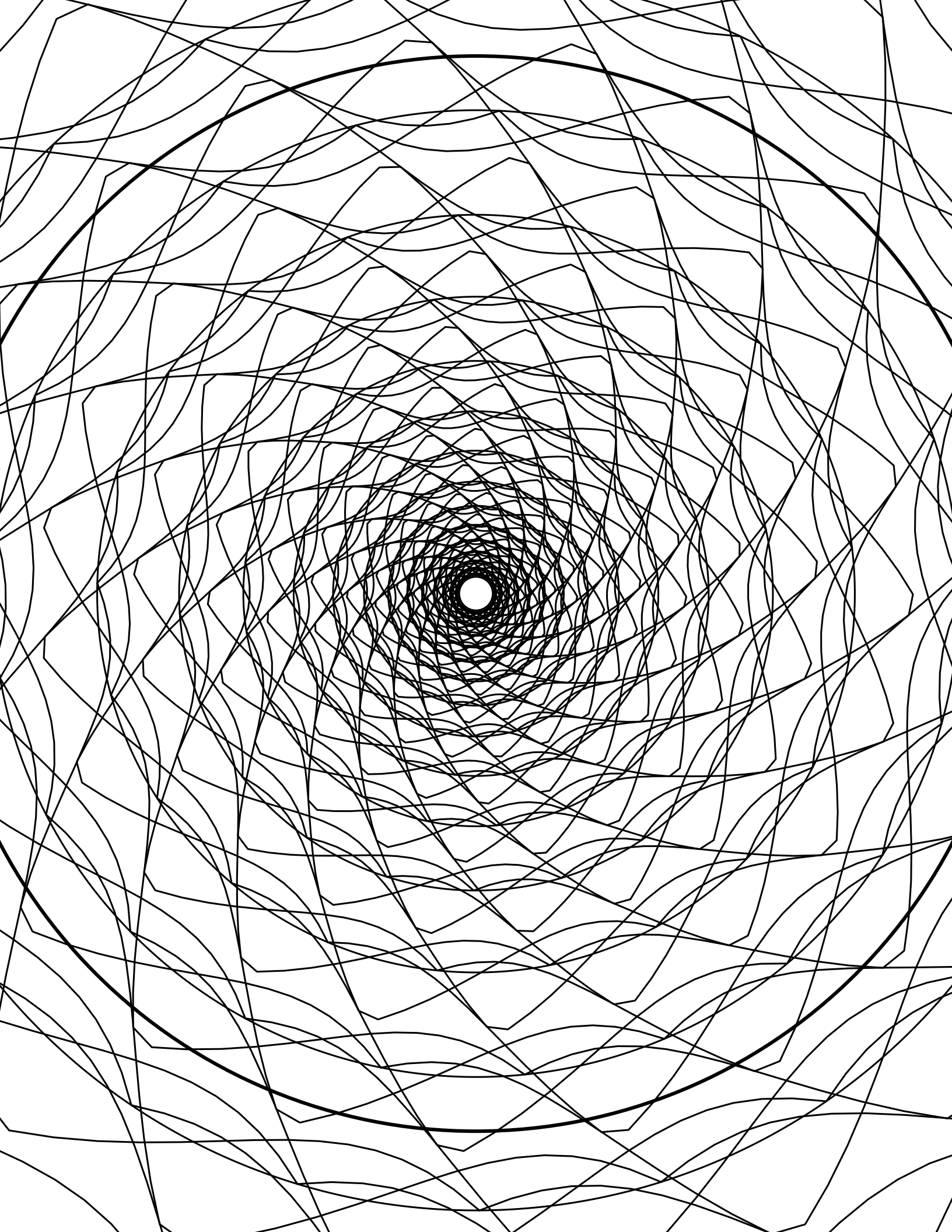


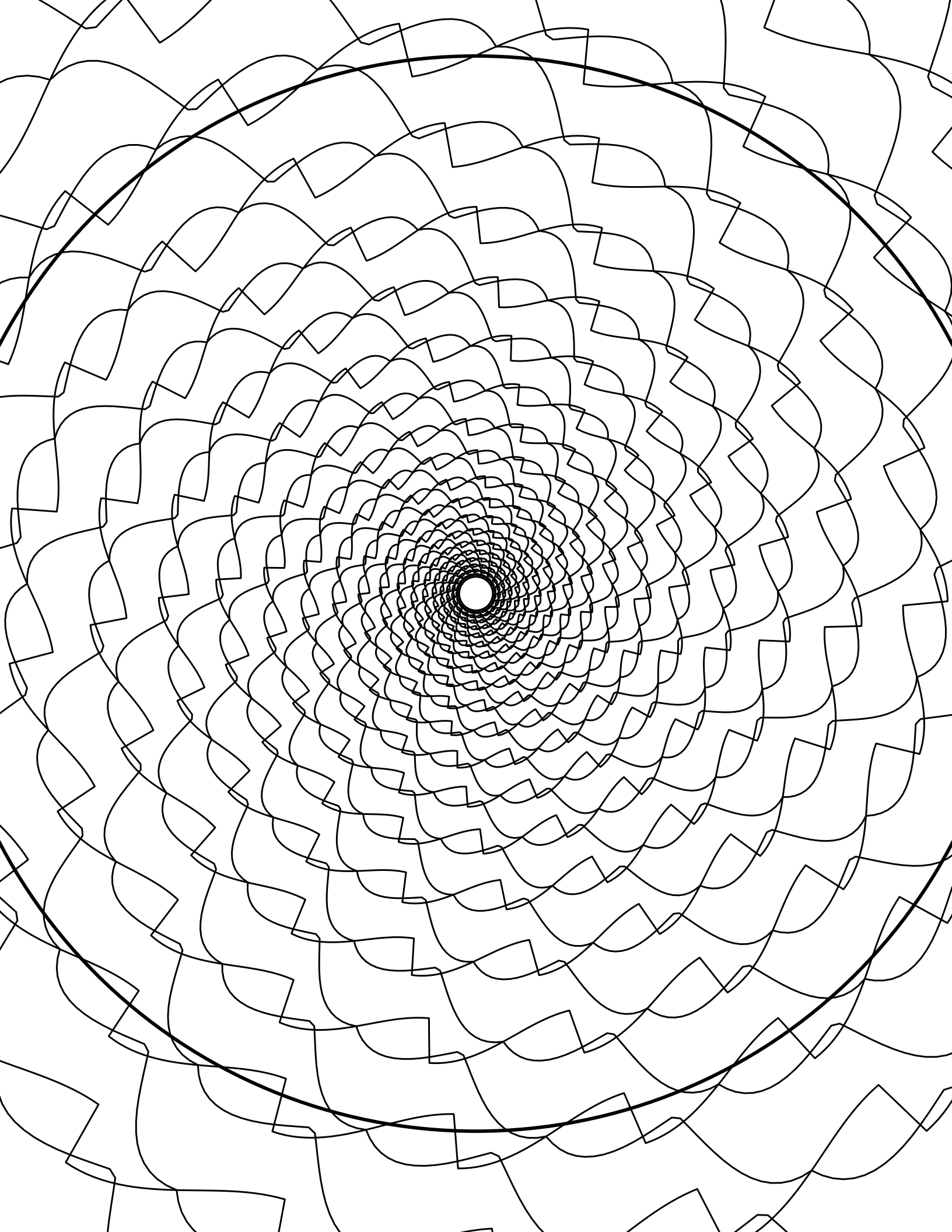


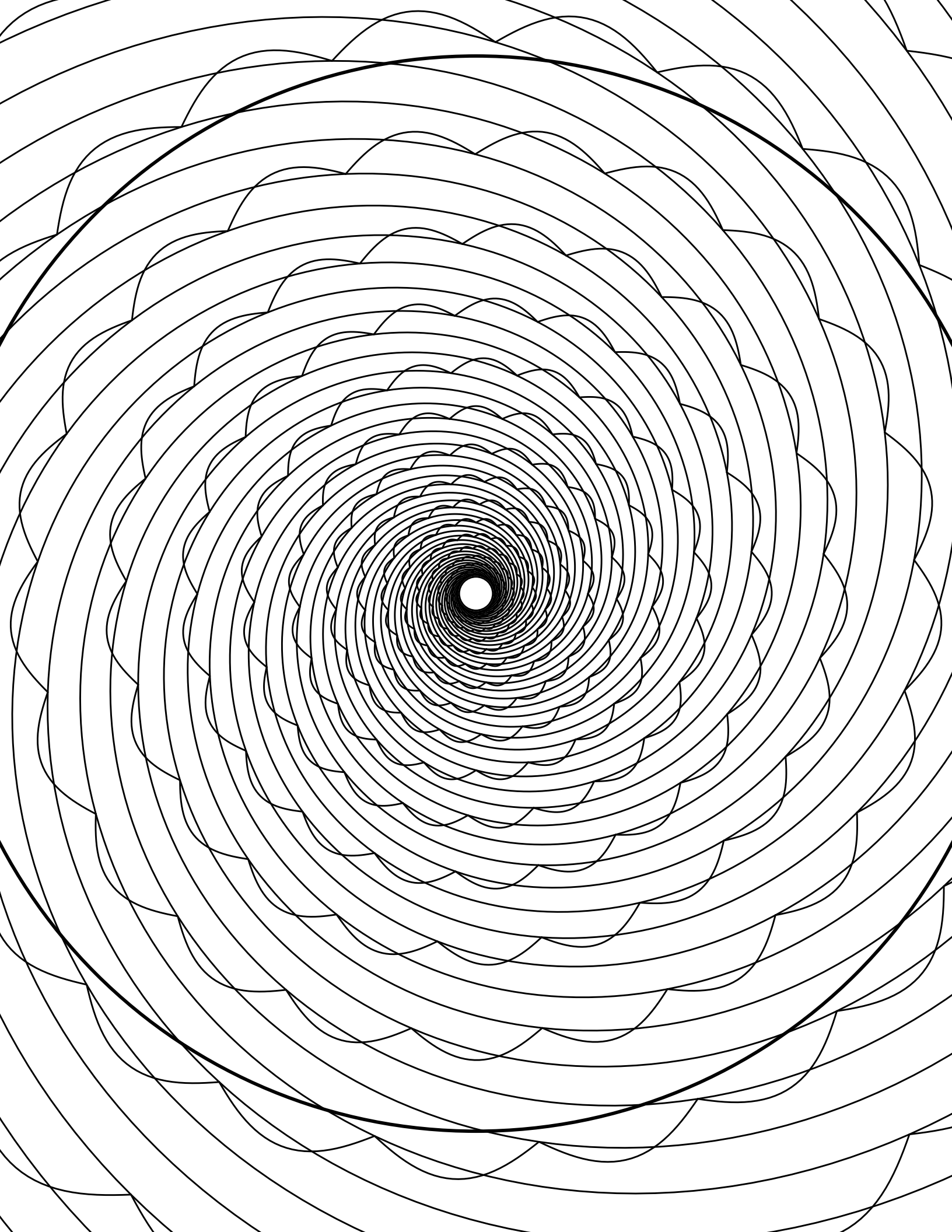


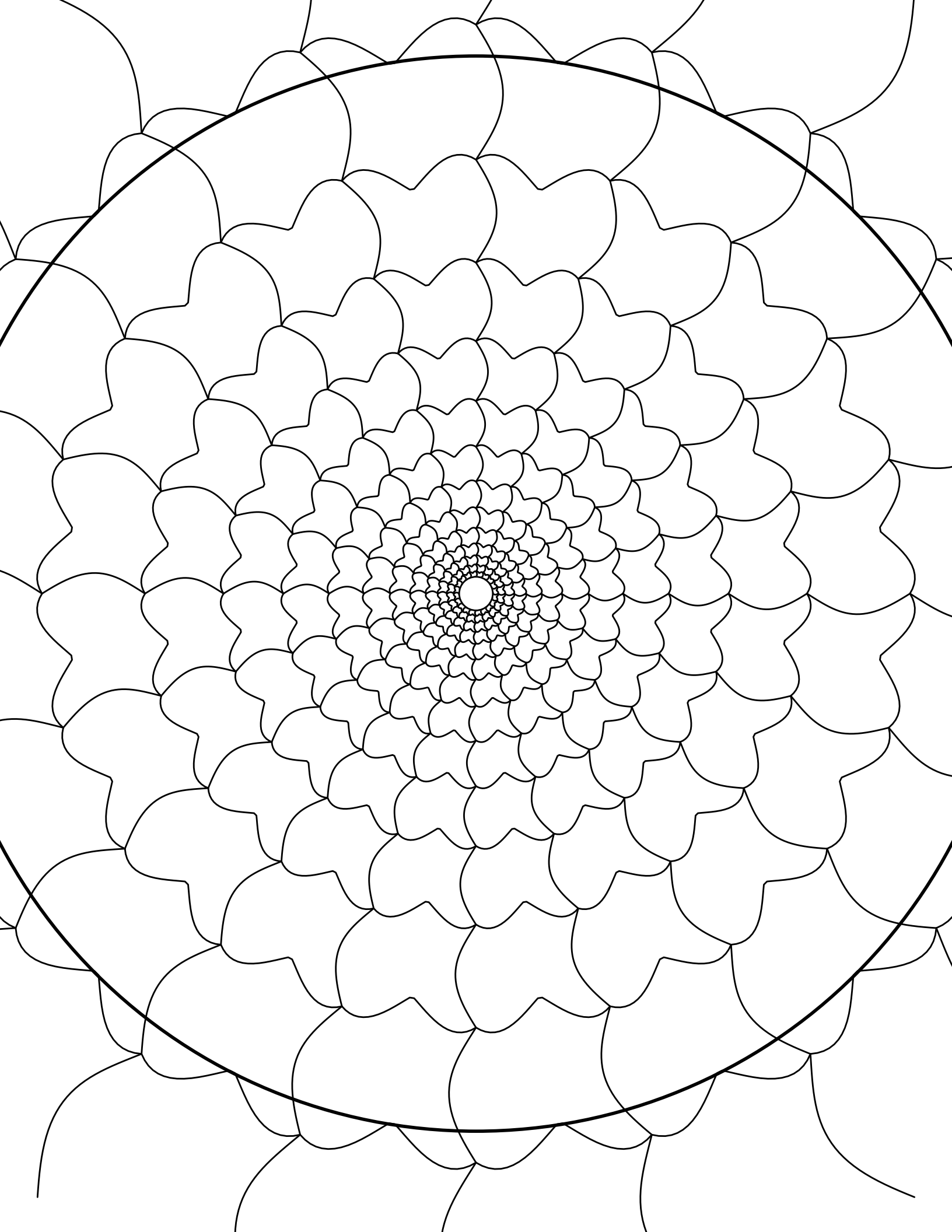


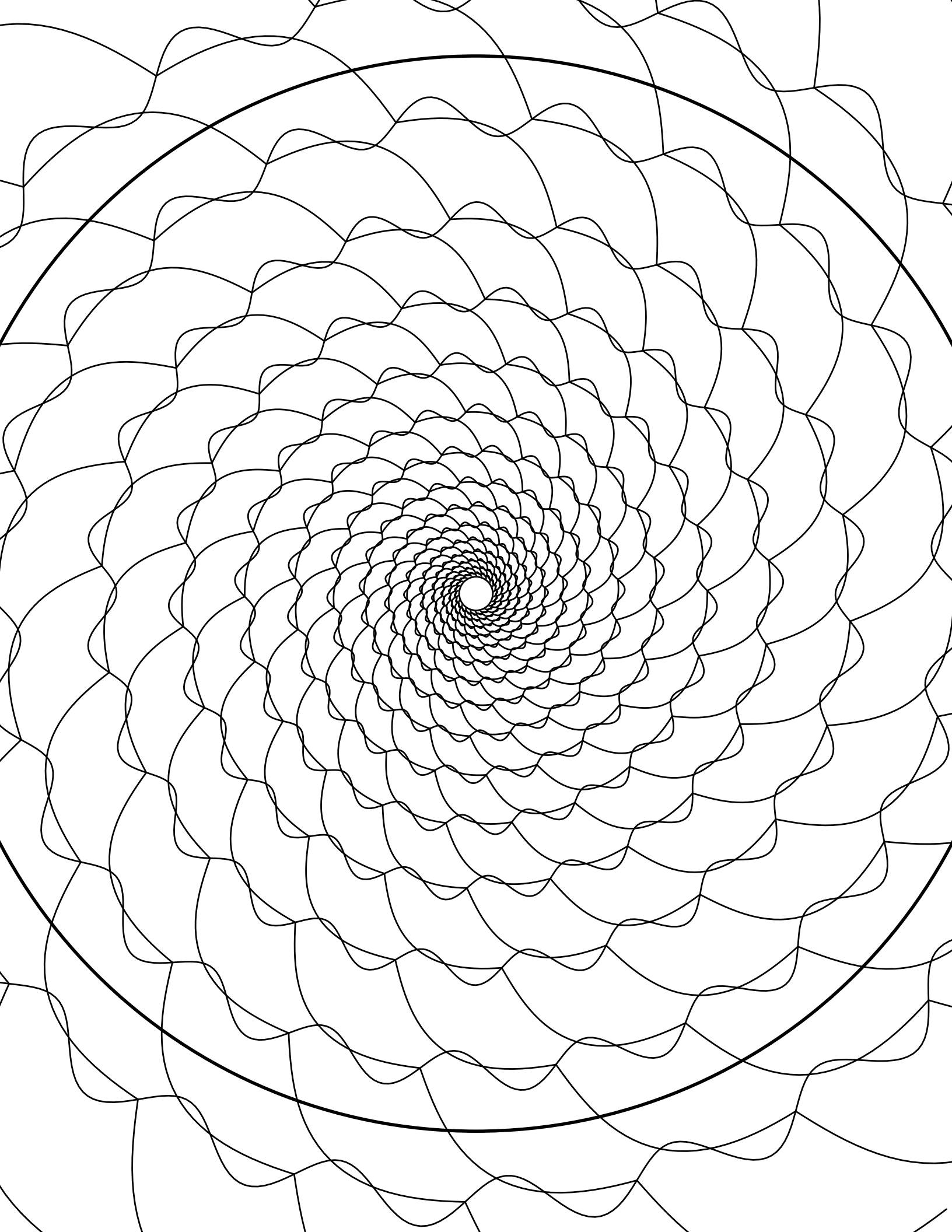


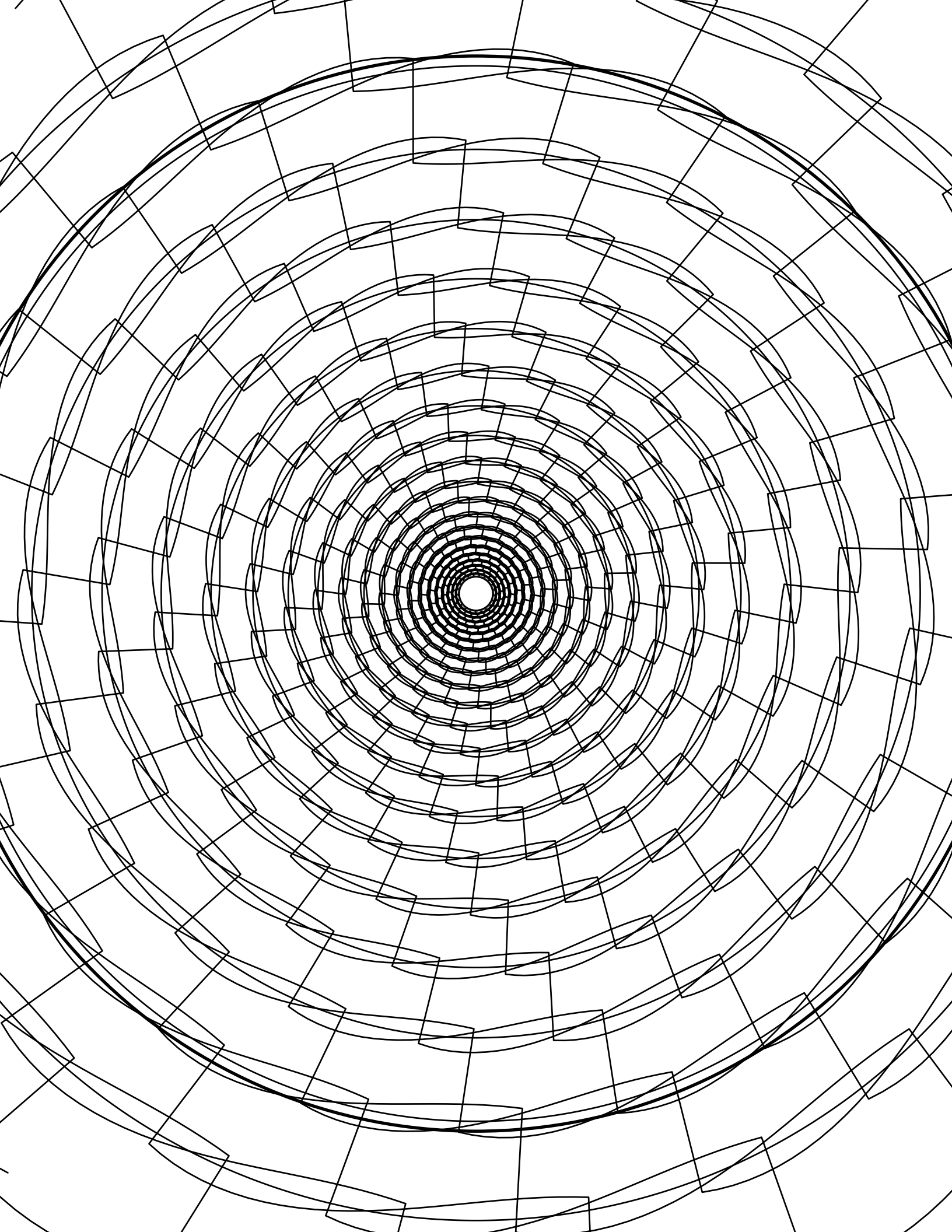


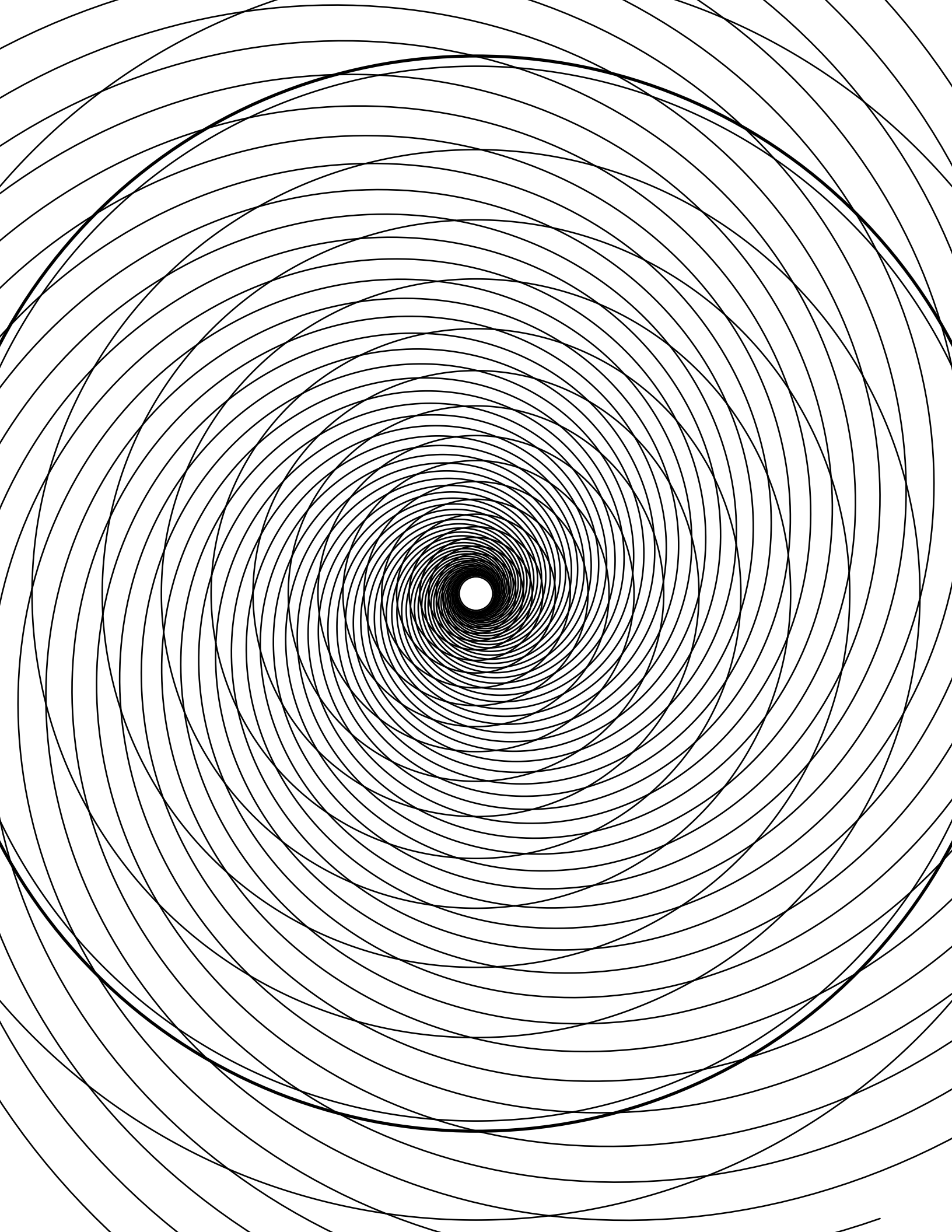


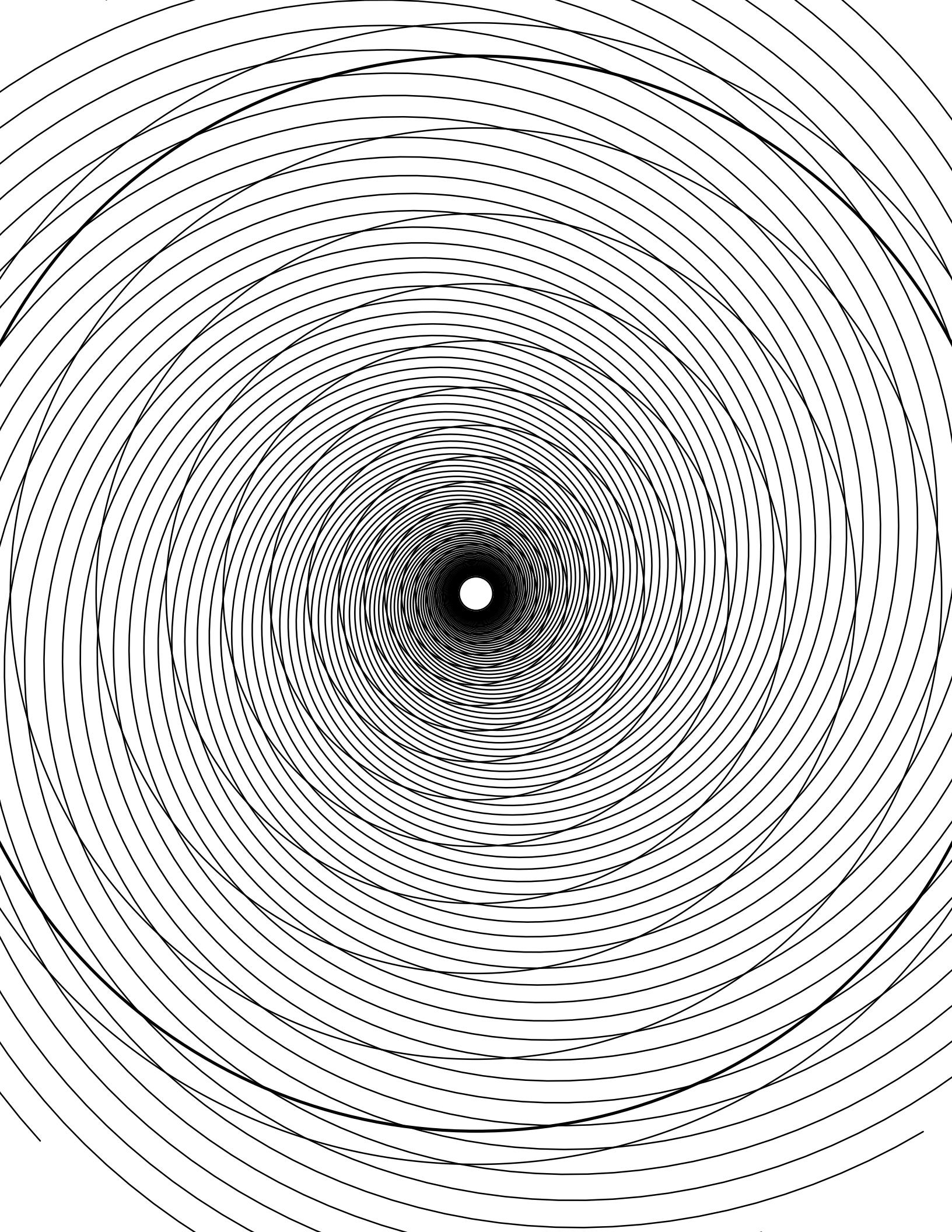


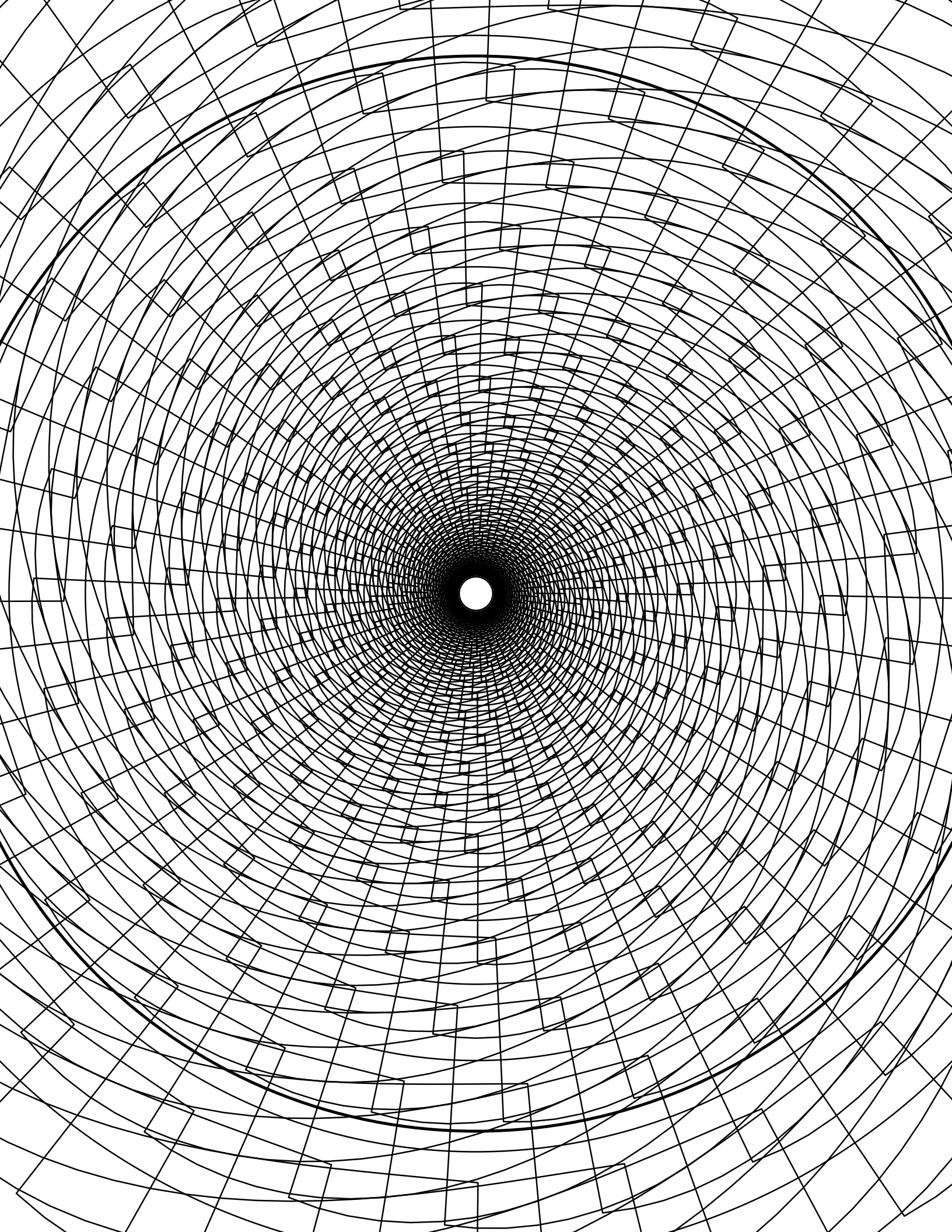


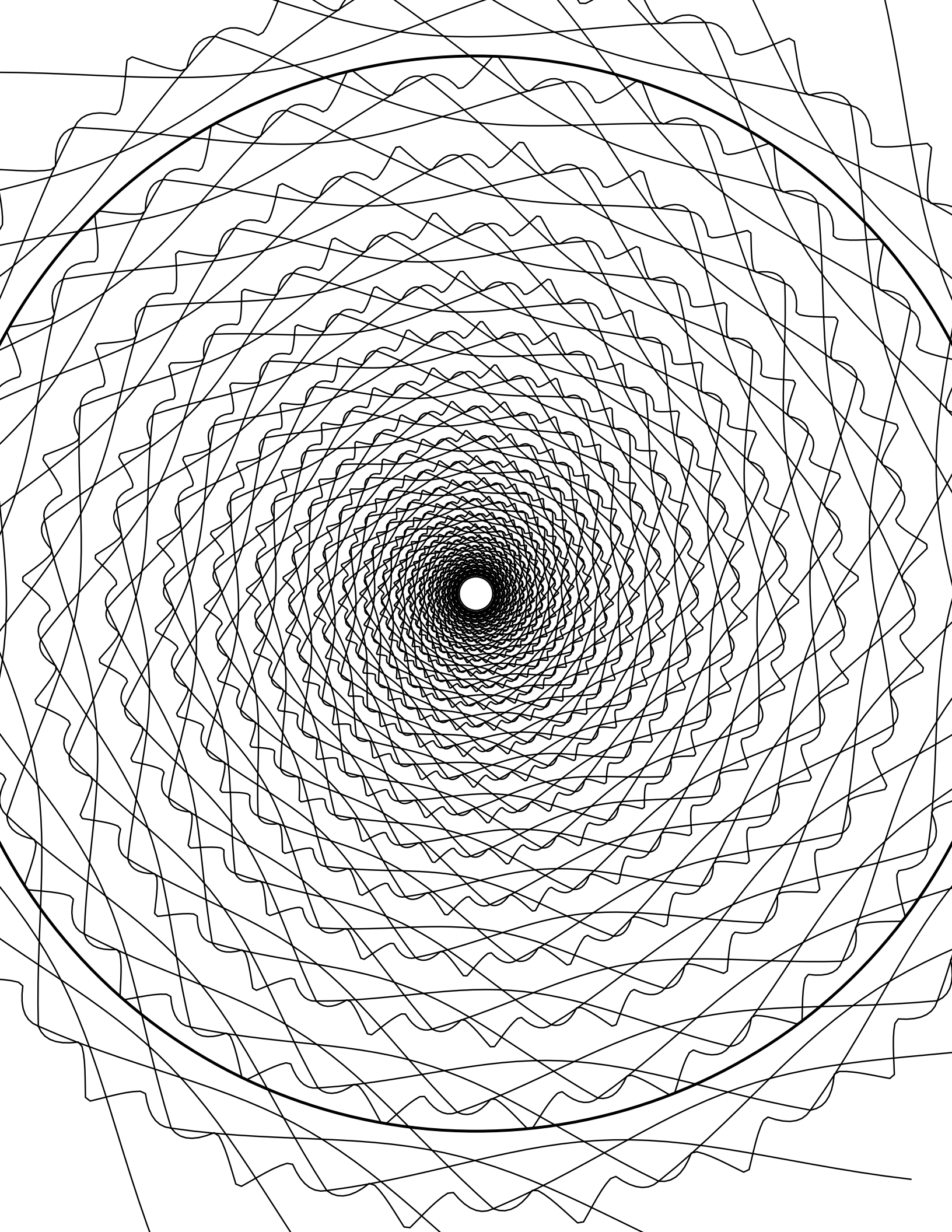


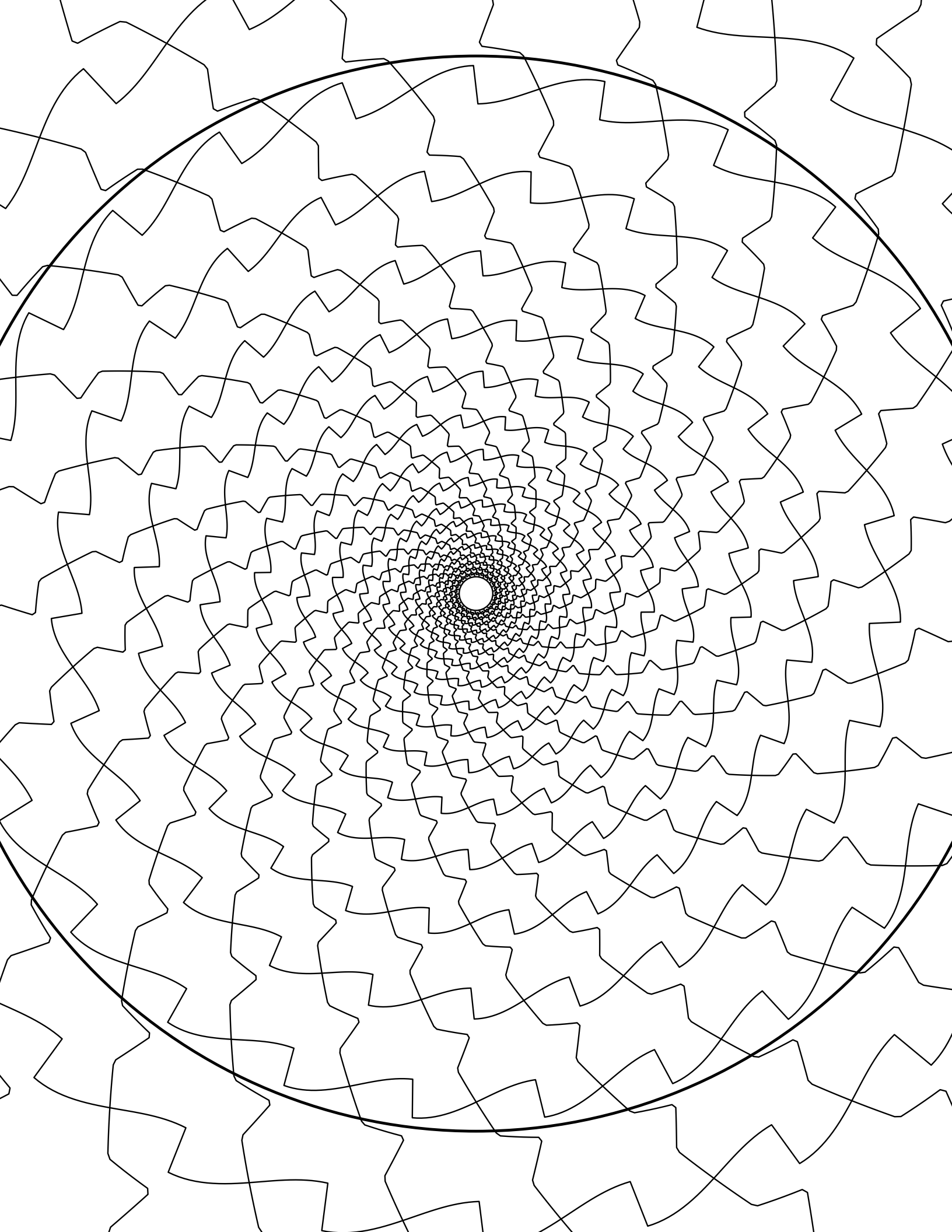


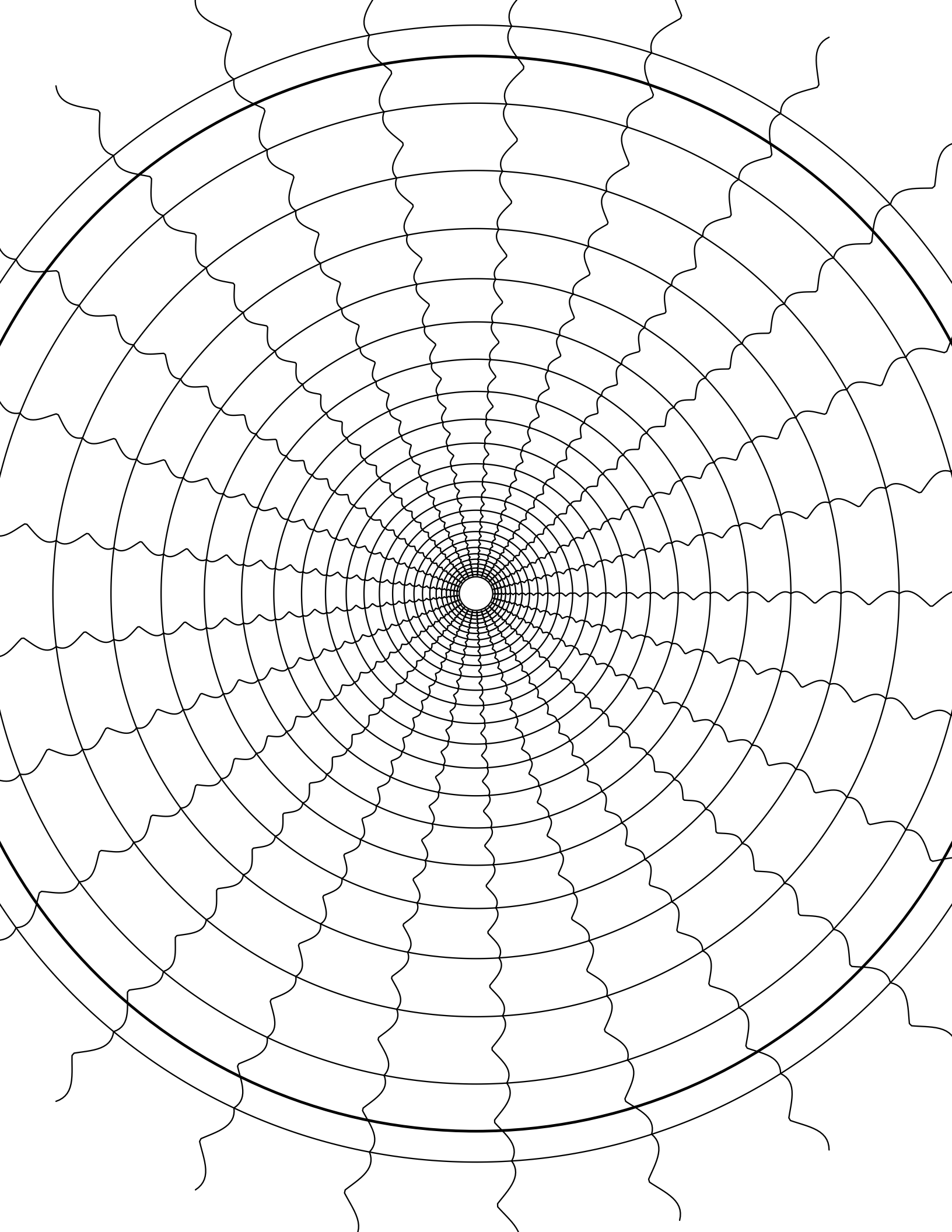


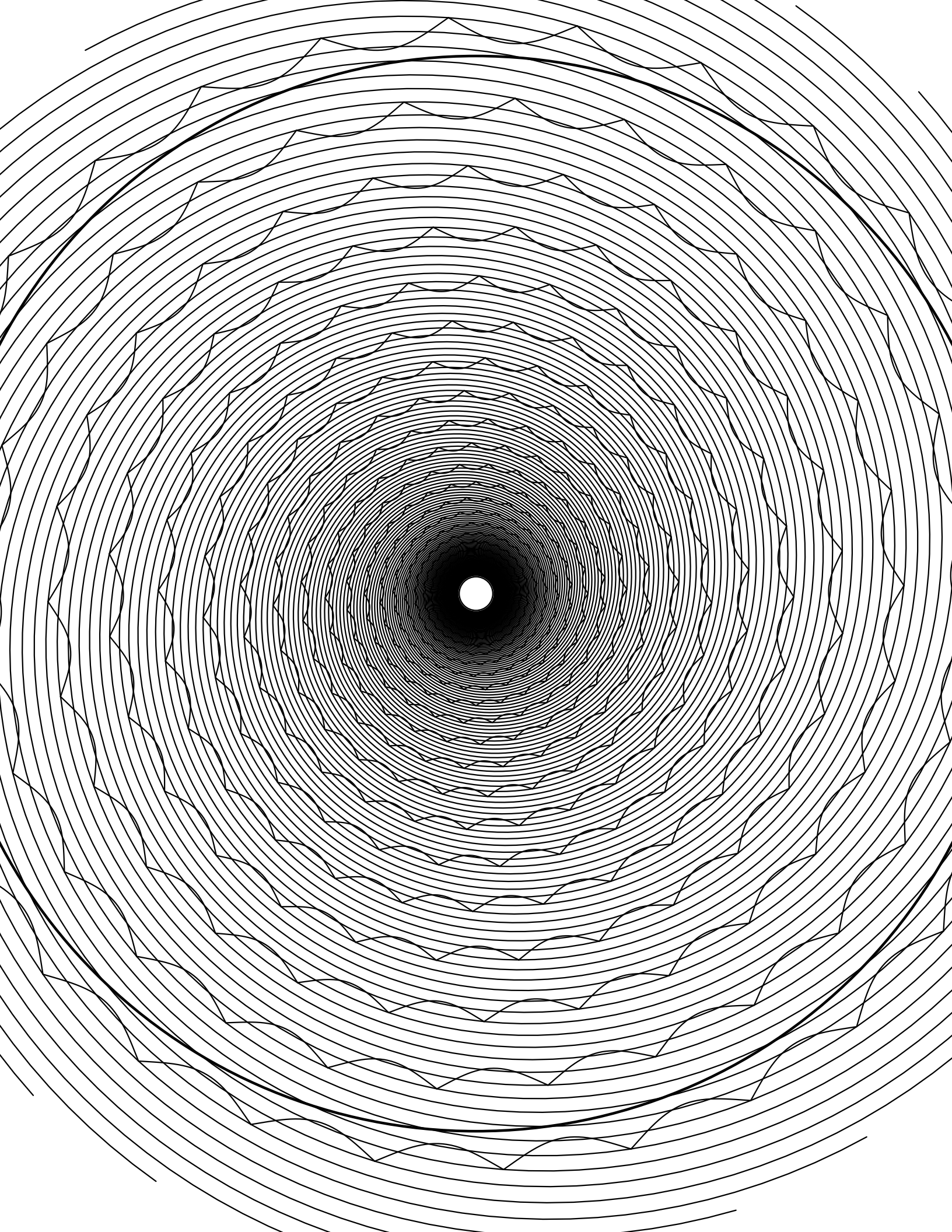


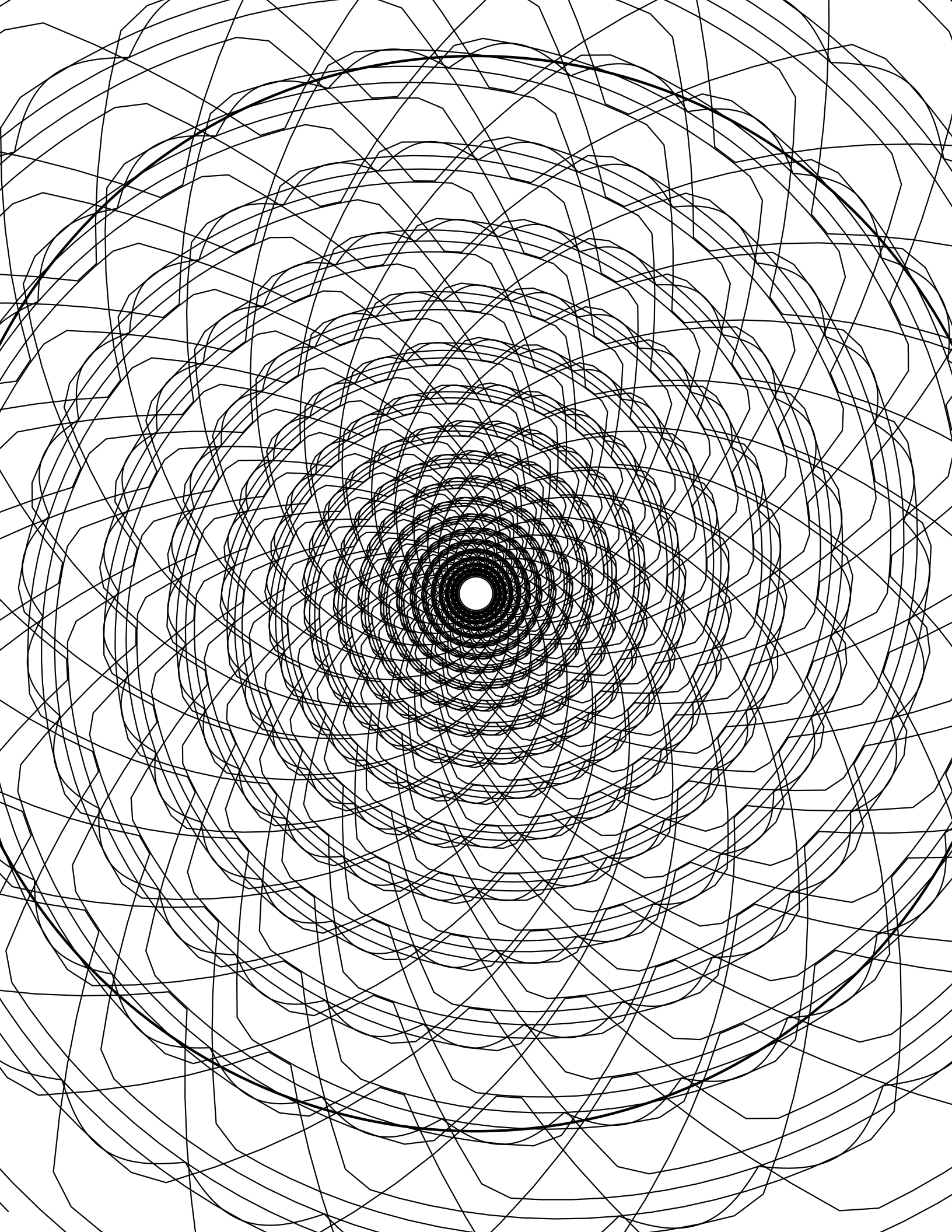


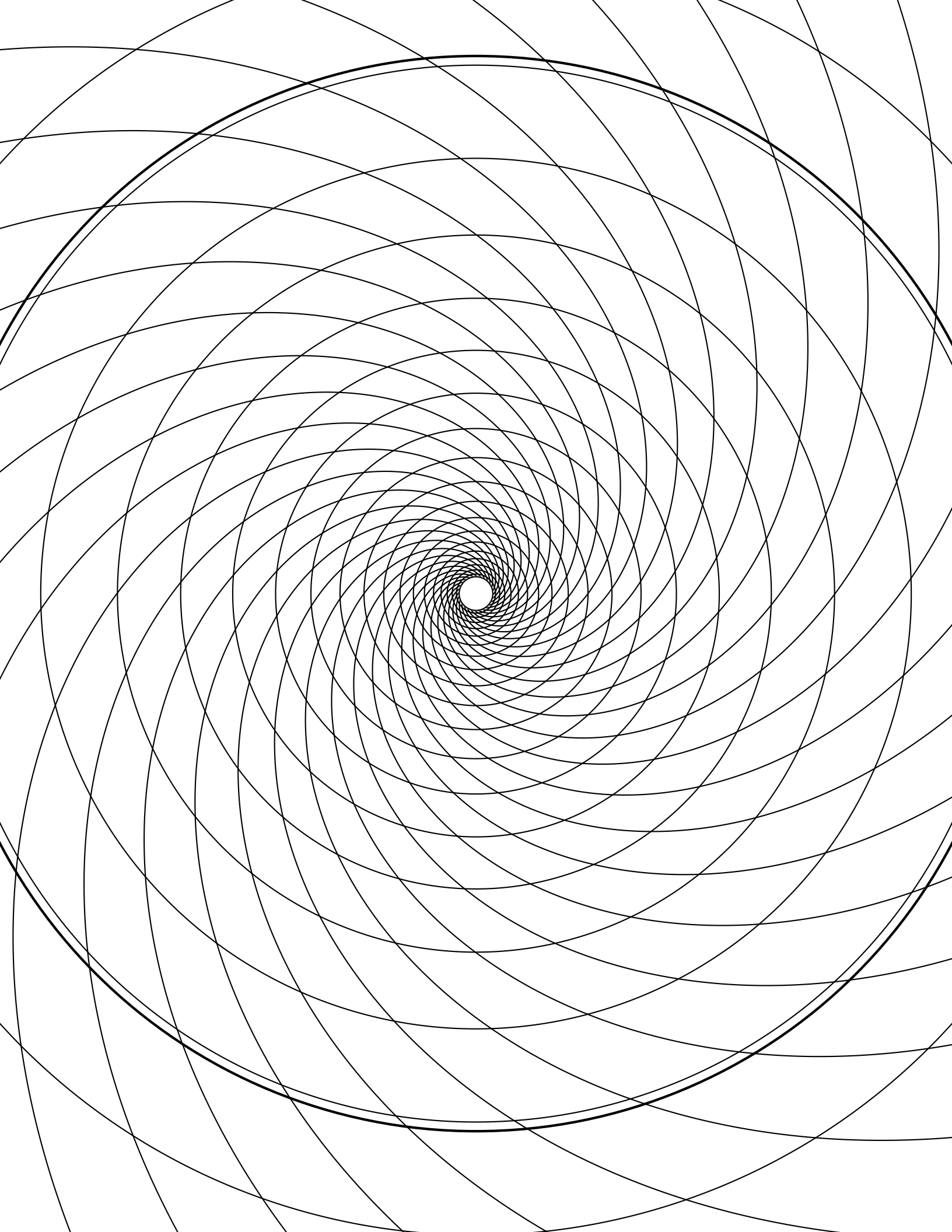












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